ECE 111: Advanced Digital Design Project 2024

The Viterbi Algorithm

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Part 1: The Viterbi Algorithm

Viterbi_tx_rx

```
module viterbi_tx_rx #(parameter N=2) (
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;  // connects encoder to decoder

    int           error_counter,
                  bad_bit_ct,
                  word_ct;
    logic   [1:0] encoder_o_reg;
    logic         encoder_i_reg;
    logic         enable_decoder_in;
    logic         enable_encoder_i_reg;
    wire          valid_encoder_o;
    logic   [1:0] err_inj;

    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter         <= 'd0;
          encoder_o_reg         <= 'b0;
          enable_decoder_in     <= 'b0;
          enable_encoder_i_reg  <= 'b0;
          word_ct               <= 'b0;
       end
       else  begin
          enable_encoder_i_reg <= enable_encoder_i;
          enable_decoder_in    <= valid_encoder_o;
//          encoder_o_reg         <= 'b0;
          //error_counter        <= $random;
          word_ct               <= word_ct + 1;
// bit error injection in encoder_o_reg
          encoder_i_reg     <= encoder_i;
//          if(error_counter[N-1:0]=='1) begin
//          err_inj        <= error_counter[29:28];
//             if(word_ct<256)
//                bad_bit_ct     <= bad_bit_ct + error_counter[29] + error_counter[28];
//       $display("error_counter,err_inj = %h %b %d",error_counter,err_inj,bad_bit_ct);
//          encoder_o_reg  <= encoder_o^err_inj;  // inject bad bits
//          end
//          else begin              // clean version
            encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
            err_inj        <= 2'b0;
//    end
       end

 // insert your convolutional encoder here
 // change port names and module name as necessary/desired
   encoder encoder1       (
       .clk,
       .rst,
       .enable_i(enable_encoder_i), //_reg),
       .d_in    (encoder_i),        //_reg),
       .valid_o (valid_encoder_o),
       .d_out   (encoder_o)   );

 // insert your term project code here
   decoder decoder1       (
       .clk,
       .rst,
       .enable (enable_decoder_in),
       .d_in   (encoder_o_reg),
       .d_out  (decoder_o)   );
```


```
       end

 // insert your convolutional encoder here
 // change port names and module name as necessary/desired
   encoder encoder1       (
       .clk,
       .rst,
       .enable_i(enable_encoder_i), //_reg),
       .d_in    (encoder_i),        //_reg),
       .valid_o (valid_encoder_o),
       .d_out   (encoder_o)   );

 // insert your term project code here
   decoder decoder1       (
       .clk,
       .rst,
       .enable (enable_decoder_in),
       .d_in   (encoder_o_reg),
       .d_out  (decoder_o)   );
```


```
endmodule
```

Encoder

```
module encoder                          // use this one
(   input               clk,
    input               rst,
    input               enable_i,
    input               d_in,
    output logic        valid_o,
    output logic [1:0] d_out);

    logic           [2:0] cstate;
    logic           [2:0] nstate;
    logic                 valid_oQ;
    logic           [1:0] d_out_reg;

    always_comb begin
       valid_oQ  =   enable_i;
       case (cstate)
          3'b000: if(!d_in)  begin
                nstate     =  3'b000;
                d_out_reg =  2'b00;
             end
             else begin
                nstate     =  3'b100;
                d_out_reg =  2'b11;
             end
          3'b001: if(!d_in)  begin
                nstate     =  3'b100;
                d_out_reg =  2'b00;
             end
             else begin
                nstate     =  3'b000;
                d_out_reg =  2'b11;
             end
          3'b010: if(!d_in)  begin
                nstate     =  3'b101;
                d_out_reg =  2'b10;
             end
             else begin
                nstate     =  3'b001;
                d_out_reg =  2'b01;
             end
          3'b011:  if(!d_in) begin
                nstate     =  3'b001;
                d_out_reg =  2'b10;
             end
             else begin
                nstate     =  3'b101;
                d_out_reg =  2'b01;
             end
          3'b100:  if(!d_in)   begin
                nstate     =  3'b010;
                d_out_reg =  2'b10;
             end
             else               begin
                nstate     =  3'b110;
                d_out_reg =  2'b01;
             end
          3'b101:  if(!d_in)  begin
                nstate     =  3'b110;
                d_out_reg =  2'b10;
             end
             else       begin
                nstate     =  3'b010;
                d_out_reg =  2'b01;
             end
```

```
61              else        begin
62                 nstate     =  3'b010;
63                 d_out_reg =  2'b01;
64              end
65           3'b110:  if(!d_in)   begin
66                 nstate     =  3'b111;
67                 d_out_reg =  2'b00;
68              end
69              else     begin
70                 nstate     =  3'b011;
71                 d_out_reg =  2'b11;
72              end
73           3'b111:  if(!d_in)   begin
74                 nstate     =  3'b011;
75                 d_out_reg =  2'b00;
76              end
77              else      begin
78                 nstate     =  3'b111;
79                 d_out_reg =  2'b11;
80              end
81        endcase
82     end
83
84       always @ (posedge clk, negedge rst)
85       if(!rst)
86         cstate    <= 'b0;
87       else
88         cstate    <= enable_i? nstate : 'b0;
89     always @ (posedge clk) begin
90       d_out       <= enable_i? d_out_reg : 'b0;
91       valid_o     <= valid_oQ;
92     end
93
94   endmodule
95
```

Decoder

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```
module decoder
(
    input           clk,
    input           rst,
    input           enable,
    input [1:0]     d_in,
    output reg     d_out);

//   logic            decoder_o_reg;

//bmc module signals
    wire  [1:0]       bmc000_path_0_bmc;
    wire  [1:0]       bmc001_path_0_bmc;
    wire  [1:0]       bmc010_path_0_bmc;
    wire  [1:0]       bmc011_path_0_bmc;
    wire  [1:0]       bmc100_path_0_bmc;
    wire  [1:0]       bmc101_path_0_bmc;
    wire  [1:0]       bmc110_path_0_bmc;
    wire  [1:0]       bmc111_path_0_bmc;

    wire  [1:0]       bmc000_path_1_bmc;
    wire  [1:0]       bmc001_path_1_bmc;
    wire  [1:0]       bmc010_path_1_bmc;
    wire  [1:0]       bmc011_path_1_bmc;
    wire  [1:0]       bmc100_path_1_bmc;
    wire  [1:0]       bmc101_path_1_bmc;
    wire  [1:0]       bmc110_path_1_bmc;
    wire  [1:0]       bmc111_path_1_bmc;

//ACS modules signals
    logic   [7:0]       validity;
    logic   [7:0]       selection;
    logic   [7:0]       path_cost    [8];
    wire    [7:0]       validity_nets;
    wire    [7:0]       selection_nets;

    wire              ACS000_selection;
    wire              ACS001_selection;
    wire              ACS010_selection;
    wire              ACS011_selection;
    wire              ACS100_selection;
    wire              ACS101_selection;
    wire              ACS110_selection;
    wire              ACS111_selection;

    wire              ACS000_valid_o;
    wire              ACS001_valid_o;
    wire              ACS010_valid_o;
    wire              ACS011_valid_o;
    wire              ACS100_valid_o;
    wire              ACS101_valid_o;
    wire              ACS110_valid_o;
    wire              ACS111_valid_o;

    wire  [7:0]       ACS000_path_cost;
    wire  [7:0]       ACS001_path_cost;
    wire  [7:0]       ACS010_path_cost;
    wire  [7:0]       ACS011_path_cost;
    wire  [7:0]       ACS100_path_cost;
    wire  [7:0]       ACS101_path_cost;
    wire  [7:0]       ACS110_path_cost;
    wire  [7:0]       ACS111_path_cost;

//Trelis memory write operation
    logic   [1:0]       mem_bank;
    logic   [1:0]       mem_bank_buf;
    logic   [1:0]       mem_bank_buf_buf;
    logic               mem_bank_buf_buf_buf;
    logic               mem_bank_buf_buf_buf_buf;
    logic               mem_bank_buf_buf_buf_buf_buf;
    logic   [9:0]       wr_mem_counter;
```

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```
    logic                   mem_bank_buf_buf_buf_buf;
    logic   [9:0]           wr_mem_counter;
    logic   [9:0]           rd_mem_counter;

    logic   [9:0]           addr_mem_A;
    logic   [9:0]           addr_mem_B;
    logic   [9:0]           addr_mem_C;
    logic   [9:0]           addr_mem_D;

    logic                   wr_mem_A;
    logic                   wr_mem_B;
    logic                   wr_mem_C;
    logic                   wr_mem_D;

    logic   [7:0]           d_in_mem_A;
    logic   [7:0]           d_in_mem_B;
    logic   [7:0]           d_in_mem_C;
    logic   [7:0]           d_in_mem_D;

    wire    [7:0]           d_o_mem_A;
    wire    [7:0]           d_o_mem_B;
    wire    [7:0]           d_o_mem_C;
    wire    [7:0]           d_o_mem_D;

//Trace back module signals
    logic                   selection_tbu_0;
    logic                   selection_tbu_1;

    logic   [7:0]           d_in_0_tbu_0;
    logic   [7:0]           d_in_1_tbu_0;
    logic   [7:0]           d_in_0_tbu_1;
    logic   [7:0]           d_in_1_tbu_1;

    wire                    d_o_tbu_0;
    wire                    d_o_tbu_1;

    logic                   enable_tbu_0;
    logic                   enable_tbu_1;

//Display memory operations
    wire                    wr_disp_mem_0;
    wire                    wr_disp_mem_1;

    wire                    d_in_disp_mem_0;
    wire                    d_in_disp_mem_1;

    logic   [9:0]           wr_mem_counter_disp;
    logic   [9:0]           rd_mem_counter_disp;

    logic   [9:0]           addr_disp_mem_0;
    logic   [9:0]           addr_disp_mem_1;

//Branch matrc calculation modules

    bmc000   bmc000_inst(d_in,bmc000_path_0_bmc,bmc000_path_1_bmc);
    bmc001   bmc001_inst(d_in,bmc001_path_0_bmc,bmc001_path_1_bmc);
    bmc010   bmc010_inst(d_in,bmc010_path_0_bmc,bmc010_path_1_bmc);
    bmc011   bmc011_inst(d_in,bmc011_path_0_bmc,bmc011_path_1_bmc);
    bmc100   bmc100_inst(d_in,bmc100_path_0_bmc,bmc100_path_1_bmc);
    bmc101   bmc101_inst(d_in,bmc101_path_0_bmc,bmc101_path_1_bmc);
    bmc110   bmc110_inst(d_in,bmc110_path_0_bmc,bmc110_path_1_bmc);
    bmc111   bmc111_inst(d_in,bmc111_path_0_bmc,bmc111_path_1_bmc);



//Add Compare Select Modules
    ACS      ACS000(validity[0],validity[1],bmc000_path_0_bmc,bmc000_path_1_bmc,path_cost[0],path_cost[1],ACS000_selection,ACS000_valid_o,ACS000_path_cost);
    ACS      ACS001(validity[3],validity[2],bmc001_path_0_bmc,bmc001_path_1_bmc,path_cost[3],path_cost[2],ACS001_selection,ACS001_valid_o,ACS001_path_cost);
    ACS      ACS010(validity[4],validity[5],bmc010_path_0_bmc,bmc010_path_1_bmc,path_cost[4],path_cost[5],ACS010_selection,ACS010_valid_o,ACS010_path_cost);
```

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```
    ACS      ACS001(validity[3],validity[2],bmc001_path_0_bmc,bmc001_path_1_bmc,path_cost[3],path_cost[2],ACS001_selection,ACS001_valid_o,ACS001_path_cost);
    ACS      ACS010(validity[4],validity[5],bmc010_path_0_bmc,bmc010_path_1_bmc,path_cost[4],path_cost[5],ACS010_selection,ACS010_valid_o,ACS010_path_cost);
    ACS      ACS011(validity[7],validity[6],bmc011_path_0_bmc,bmc011_path_1_bmc,path_cost[7],path_cost[6],ACS011_selection,ACS011_valid_o,ACS011_path_cost);
    ACS      ACS100(validity[1],validity[0],bmc100_path_0_bmc,bmc100_path_1_bmc,path_cost[1],path_cost[0],ACS100_selection,ACS100_valid_o,ACS100_path_cost);
    ACS      ACS101(validity[2],validity[3],bmc101_path_0_bmc,bmc101_path_1_bmc,path_cost[2],path_cost[3],ACS101_selection,ACS101_valid_o,ACS101_path_cost);
    ACS      ACS110(validity[5],validity[4],bmc110_path_0_bmc,bmc110_path_1_bmc,path_cost[5],path_cost[4],ACS110_selection,ACS110_valid_o,ACS110_path_cost);
    ACS      ACS111(validity[6],validity[7],bmc111_path_0_bmc,bmc111_path_1_bmc,path_cost[6],path_cost[7],ACS111_selection,ACS111_valid_o,ACS111_path_cost);

    assign selection_nets  =  {ACS111_selection,ACS110_selection,ACS101_selection,ACS100_selection,
                               ACS011_selection,ACS010_selection,ACS001_selection,ACS000_selection};
    assign validity_nets   =  {ACS111_valid_o,ACS110_valid_o,ACS101_valid_o,ACS100_valid_o,
                               ACS011_valid_o,ACS010_valid_o,ACS001_valid_o,ACS000_valid_o};


    always @ (posedge clk, negedge rst) begin
        if(!rst)  begin
            validity            <= 8'b00000001;
            selection           <= 8'b00000000;
// clear all 8 path costs
            path_cost[0]        <= 1'b0;
            path_cost[1]        <= 1'b0;
            path_cost[2]        <= 1'b0;
            path_cost[3]        <= 1'b0;
            path_cost[4]        <= 1'b0;
            path_cost[5]        <= 1'b0;
            path_cost[6]        <= 1'b0;
            path_cost[7]        <= 1'b0;

        end
        else if(!enable)   begin
            validity            <= 8'b00000001;
            selection           <= 8'b00000000;
// clear all 8 path costs
            path_cost[0]        <= 1'b0;
            path_cost[1]        <= 1'b0;
            path_cost[2]        <= 1'b0;
            path_cost[3]        <= 1'b0;
            path_cost[4]        <= 1'b0;
            path_cost[5]        <= 1'b0;
            path_cost[6]        <= 1'b0;
            path_cost[7]        <= 1'b0;

        end
        else if( path_cost[0][7] && path_cost[1][7] && path_cost[2][7] && path_cost[3][7] &&
                 path_cost[4][7] && path_cost[5][7] && path_cost[6][7] && path_cost[7][7] )
        begin

            validity            <= validity_nets;
            selection           <= selection_nets;

            path_cost[0]        <= 8'b01111111 & ACS000_path_cost;
            path_cost[1]        <= 8'b01111111 & ACS001_path_cost;
            path_cost[2]        <= 8'b01111111 & ACS010_path_cost;
            path_cost[3]        <= 8'b01111111 & ACS011_path_cost;
            path_cost[4]        <= 8'b01111111 & ACS100_path_cost;
            path_cost[5]        <= 8'b01111111 & ACS101_path_cost;
            path_cost[6]        <= 8'b01111111 & ACS110_path_cost;
            path_cost[7]        <= 8'b01111111 & ACS111_path_cost;
/*  likewise for path_cost[1:7] and ACS001:111_path_cost DONE
*/
        end
        else   begin
            validity            <= validity_nets;
            selection           <= selection_nets;

            path_cost[0]        <= ACS000_path_cost;
            path_cost[1]        <= ACS001_path_cost;
            path_cost[2]        <= ACS010_path_cost;
            path_cost[3]        <= ACS011_path_cost;
            path_cost[4]        <= ACS100_path_cost;
            path_cost[5]        <= ACS101_path_cost;
```

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```
    ACS       ACS001(validity[1],validity[2],bmc001_path_0_bmc,bmc001_path_1_bmc,path_cost[3],path_cost[2],ACS001_selection,ACS001_valid_o,ACS001_path_cost);
    ACS       ACS010(validity[4],validity[5],bmc010_path_0_bmc,bmc010_path_1_bmc,path_cost[4],path_cost[5],ACS010_selection,ACS010_valid_o,ACS010_path_cost);
    ACS       ACS011(validity[7],validity[6],bmc011_path_0_bmc,bmc011_path_1_bmc,path_cost[7],path_cost[6],ACS011_selection,ACS011_valid_o,ACS011_path_cost);
    ACS       ACS100(validity[1],validity[0],bmc100_path_0_bmc,bmc100_path_1_bmc,path_cost[1],path_cost[0],ACS100_selection,ACS100_valid_o,ACS100_path_cost);
    ACS       ACS101(validity[2],validity[3],bmc101_path_0_bmc,bmc101_path_1_bmc,path_cost[2],path_cost[3],ACS101_selection,ACS101_valid_o,ACS101_path_cost);
    ACS       ACS110(validity[5],validity[4],bmc110_path_0_bmc,bmc110_path_1_bmc,path_cost[5],path_cost[4],ACS110_selection,ACS110_valid_o,ACS110_path_cost);
    ACS       ACS111(validity[6],validity[7],bmc111_path_0_bmc,bmc111_path_1_bmc,path_cost[6],path_cost[7],ACS111_selection,ACS111_valid_o,ACS111_path_cost);

    assign selection_nets  =  {ACS111_selection,ACS110_selection,ACS101_selection,ACS100_selection,
                               ACS011_selection,ACS010_selection,ACS001_selection,ACS000_selection};
    assign validity_nets   =  {ACS111_valid_o,ACS110_valid_o,ACS101_valid_o,ACS100_valid_o,
                               ACS011_valid_o,ACS010_valid_o,ACS001_valid_o,ACS000_valid_o};


    always @ (posedge clk, negedge rst) begin
       if(!rst)  begin
          validity          <= 8'b00000001;
          selection         <= 8'b00000000;
 // clear all 8 path costs
          path_cost[0]      <= 1'b0;
          path_cost[1]      <= 1'b0;
          path_cost[2]      <= 1'b0;
          path_cost[3]      <= 1'b0;
          path_cost[4]      <= 1'b0;
          path_cost[5]      <= 1'b0;
          path_cost[6]      <= 1'b0;
          path_cost[7]      <= 1'b0;

       end
       else if(!enable)   begin
          validity          <= 8'b00000001;
          selection         <= 8'b00000000;
 // clear all 8 path costs
          path_cost[0]      <= 1'b0;
          path_cost[1]      <= 1'b0;
          path_cost[2]      <= 1'b0;
          path_cost[3]      <= 1'b0;
          path_cost[4]      <= 1'b0;
          path_cost[5]      <= 1'b0;
          path_cost[6]      <= 1'b0;
          path_cost[7]      <= 1'b0;

       end
       else if( path_cost[0][7] && path_cost[1][7] && path_cost[2][7] && path_cost[3][7] &&
                path_cost[4][7] && path_cost[5][7] && path_cost[6][7] && path_cost[7][7] )
       begin

          validity          <= validity_nets;
          selection         <= selection_nets;

          path_cost[0]      <= 8'b01111111 & ACS000_path_cost;
          path_cost[1]      <= 8'b01111111 & ACS001_path_cost;
          path_cost[2]      <= 8'b01111111 & ACS010_path_cost;
          path_cost[3]      <= 8'b01111111 & ACS011_path_cost;
          path_cost[4]      <= 8'b01111111 & ACS100_path_cost;
          path_cost[5]      <= 8'b01111111 & ACS101_path_cost;
          path_cost[6]      <= 8'b01111111 & ACS110_path_cost;
          path_cost[7]      <= 8'b01111111 & ACS111_path_cost;
 /*  likewise for path_cost[1:7] and ACS001:111_path_cost DONE
 */
       end
       else   begin
          validity          <= validity_nets;
          selection         <= selection_nets;

          path_cost[0]      <= ACS000_path_cost;
          path_cost[1]      <= ACS001_path_cost;
          path_cost[2]      <= ACS010_path_cost;
          path_cost[3]      <= ACS011_path_cost;
          path_cost[4]      <= ACS100_path_cost;
          path_cost[5]      <= ACS101_path_cost;
```

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```
          path_cost[4]      <= ACS100_path_cost;
          path_cost[5]      <= ACS101_path_cost;
          path_cost[6]      <= ACS110_path_cost;
          path_cost[7]      <= ACS111_path_cost;
 /* likewise for 1:7 DONE
 */

       end
    end

    always @ (posedge clk, negedge rst) begin
       if(!rst)
          wr_mem_counter <= 10'd0;
       else if(!enable)
          wr_mem_counter <= 10'd0;
       else
          wr_mem_counter <= wr_mem_counter + 10'd1;
    end

    always @ (posedge clk, negedge rst) begin
       if(!rst)
          rd_mem_counter <= 10'b1111111111;// -1   how do you handle this in 10 bit binary?
       else if(enable)
          rd_mem_counter <= rd_mem_counter - 10'd1;
    end

    always @ (posedge clk, negedge rst)
       if(!rst)
          mem_bank <= 2'b00;
       else begin
          if(wr_mem_counter==10'b1111111111)
             mem_bank <= mem_bank + 2'b01;
       end

    always @ (posedge clk)    begin
       d_in_mem_A  <= selection;
       d_in_mem_B  <= selection;
       d_in_mem_C  <= selection;
       d_in_mem_D  <= selection;
    end

    always @ (posedge clk)     begin
       case(mem_bank)
          2'b00:          begin
             addr_mem_A        <= wr_mem_counter;
             addr_mem_B        <= rd_mem_counter;
             addr_mem_C        <= 10'd0;
             addr_mem_D        <= rd_mem_counter;

             wr_mem_A          <= 1'b1;
             wr_mem_B          <= 1'b0;
             wr_mem_C          <= 1'b0;
             wr_mem_D          <= 1'b0;
 /* other wr_mems = 0
 */
          end
          2'b01:          begin
             addr_mem_A        <= rd_mem_counter;
             addr_mem_B        <= wr_mem_counter;
             addr_mem_C        <= rd_mem_counter;
             addr_mem_D        <= 10'd0;

             wr_mem_B          <= 1'b1;
             wr_mem_A          <= 1'b0;
             wr_mem_C          <= 1'b0;
             wr_mem_D          <= 1'b0;
 /* other wr_mems = 0
 */
          end
          2'b10:    begin
             addr_mem_A        <= 10'd0;
```

Ln 240 Col 10 SystemVerilog HDL File 100% 00:00:36

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```
        2'b10:      begin
            addr_mem_A          <= 10'd0;
            addr_mem_B          <= rd_mem_counter;
            addr_mem_C          <= wr_mem_counter;
            addr_mem_D          <= rd_mem_counter;

            wr_mem_C        <= 1'b1;
            wr_mem_A            <= 1'b0;
            wr_mem_B            <= 1'b0;
            wr_mem_D            <= 1'b0;
 /* other wr_mems = 0
 */
        end
        2'b11:      begin
            addr_mem_A          <= rd_mem_counter;
            addr_mem_B          <= 10'd0;
            addr_mem_C          <= rd_mem_counter;
            addr_mem_D          <= wr_mem_counter;

            wr_mem_D            <= 1'b1;
            wr_mem_A            <= 1'b0;
            wr_mem_B            <= 1'b0;
            wr_mem_C            <= 1'b0;
 /* other wr_mems = 0
 */
        end
      endcase
  end

 //Trelis memory module instantiation

   mem   trelis_mem_A
   (
      .clk,
      .wr(wr_mem_A),
      .addr(addr_mem_A),
      .d_i(d_in_mem_A),
      .d_o(d_o_mem_A)
   );
 /* likewise for trelis_memB, C, D
 */
   mem   trelis_mem_B
   (
      .clk,
      .wr(wr_mem_B),
      .addr(addr_mem_B),
      .d_i(d_in_mem_B),
      .d_o(d_o_mem_B)
   );
   mem   trelis_mem_C
   (
      .clk,
      .wr(wr_mem_C),
      .addr(addr_mem_C),
      .d_i(d_in_mem_C),
      .d_o(d_o_mem_C)
   );
   mem   trelis_mem_D
   (
      .clk,
      .wr(wr_mem_D),
      .addr(addr_mem_D),
      .d_i(d_in_mem_D),
      .d_o(d_o_mem_D)
   );

 //Trace back module operation

   always @(posedge clk)
      mem_bank_buf   <= mem_bank;
```

Ln 314 Col 22 SystemVerilog HDL File 100% 00:00:36

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```
      mem_bank_buf   <= mem_bank;

   always @(posedge clk)
      mem_bank_buf_buf   <= mem_bank_buf;

   always @ (posedge clk, negedge rst)
      if(!rst)
            enable_tbu_0   <= 1'b0;
      else begin
         if(mem_bank_buf_buf==2'b10)
            enable_tbu_0   <= 1'b1;
         else
            enable_tbu_0   <= enable_tbu_0;
      end

   always @ (posedge clk, negedge rst)
      if(!rst)
            enable_tbu_1   <= 1'b0;
      else begin
         if(mem_bank_buf_buf==2'b11)
            enable_tbu_1   <= 1'b1;
         else
            enable_tbu_1   <= enable_tbu_1;
      end

   always @ (posedge clk)
      case(mem_bank_buf_buf)
         2'b00:  begin
            d_in_0_tbu_0   <= d_o_mem_D;
            d_in_1_tbu_0   <= d_o_mem_C;

            d_in_0_tbu_1   <= d_o_mem_C;
            d_in_1_tbu_1   <= d_o_mem_B;

            selection_tbu_0<= 1'b0;
            selection_tbu_1<= 1'b1;

         end
         2'b01:   begin
            d_in_0_tbu_0   <= d_o_mem_D;
            d_in_1_tbu_0   <= d_o_mem_C;

            d_in_0_tbu_1   <= d_o_mem_A;
            d_in_1_tbu_1   <= d_o_mem_D;

            selection_tbu_0<= 1'b1;
            selection_tbu_1<= 1'b0;
         end
         2'b10:   begin
            d_in_0_tbu_0   <= d_o_mem_B;
            d_in_1_tbu_0   <= d_o_mem_A;

            d_in_0_tbu_1   <= d_o_mem_A;
            d_in_1_tbu_1   <= d_o_mem_D;

            selection_tbu_0<= 1'b0;
            selection_tbu_1<= 1'b1;
         end
         2'b11:  begin
            d_in_0_tbu_0   <= d_o_mem_B;
            d_in_1_tbu_0   <= d_o_mem_A;

            d_in_0_tbu_1   <= d_o_mem_C;
            d_in_1_tbu_1   <= d_o_mem_B;

            selection_tbu_0<= 1'b1;
            selection_tbu_1<= 1'b0;
         end
      endcase

 //Trace-Back modules instantiation
```

Ln 389 Col 12 SystemVerilog HDL File 100% 00:00:36

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```
 //Trace-Back modules instantiation

   tbu tbu_0   (
       .clk,
       .rst,
       .enable(enable_tbu_0),
       .selection(selection_tbu_0),
       .d_in_0(d_in_0_tbu_0),
       .d_in_1(d_in_1_tbu_0),
       .d_o(d_o_tbu_0),
       .wr_en(wr_disp_mem_0)
   );

 /* analogous for tbu_1
 */
   tbu tbu_1   (
       .clk,
       .rst,
       .enable(enable_tbu_1),
       .selection(selection_tbu_1),
       .d_in_0(d_in_0_tbu_1),
       .d_in_1(d_in_1_tbu_1),
       .d_o(d_o_tbu_1),
       .wr_en(wr_disp_mem_1)
   );
 //Display Memory modules Instantioation

    assign   d_in_disp_mem_0   =  d_o_tbu_0;
    assign   d_in_disp_mem_1   =  d_o_tbu_1;

   mem_disp   disp_mem_0
   (
       .clk
       .wr(wr_disp_mem_0),
       .addr(addr_disp_mem_0),
       .d_i(d_in_disp_mem_0),
       .d_o(d_o_disp_mem_0)
   );
 /* analogous for disp_mem_1
 */
   mem_disp   disp_mem_1
   (
       .clk
       .wr(wr_disp_mem_1),
       .addr(addr_disp_mem_1),
       .d_i(d_in_disp_mem_1),
       .d_o(d_o_disp_mem_1)
   );
 // Display memory module operation
    always @ (posedge clk)
       mem_bank_buf_buf_buf <= mem_bank_buf_buf[0];

    always @ (posedge clk)
       if(!rst)
          wr_mem_counter_disp  <= 10'b0000000010;
       else if(!enable)
          wr_mem_counter_disp  <= 10'b0000000010;
       else
          wr_mem_counter_disp  <= wr_mem_counter_disp - 10'd1;

    always @ (posedge clk)
       if(!rst)
          rd_mem_counter_disp  <= 10'b1111111101;
       else if(!enable)
          rd_mem_counter_disp  <= 10'b1111111101;
       else
          rd_mem_counter_disp  <= rd_mem_counter_disp + 10'd1;

    always @ (posedge clk)
       case(mem_bank_buf_buf_buf)
```

Ln 467 Col 1 SystemVerilog HDL File 100% 00:00:36

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```
    assign   d_in_disp_mem_1   =  d_o_tbu_1;

   mem_disp   disp_mem_0
   (
       .clk
       .wr(wr_disp_mem_0),
       .addr(addr_disp_mem_0),
       .d_i(d_in_disp_mem_0),
       .d_o(d_o_disp_mem_0)
   );
 /* analogous for disp_mem_1
 */
   mem_disp   disp_mem_1
   (
       .clk
       .wr(wr_disp_mem_1),
       .addr(addr_disp_mem_1),
       .d_i(d_in_disp_mem_1),
       .d_o(d_o_disp_mem_1)
   );
 // Display memory module operation
    always @ (posedge clk)
       mem_bank_buf_buf_buf <= mem_bank_buf_buf[0];

    always @ (posedge clk)
       if(!rst)
          wr_mem_counter_disp  <= 10'b0000000010;
       else if(!enable)
          wr_mem_counter_disp  <= 10'b0000000010;
       else
          wr_mem_counter_disp  <= wr_mem_counter_disp - 10'd1;

    always @ (posedge clk)
       if(!rst)
          rd_mem_counter_disp  <= 10'b1111111101;
       else if(!enable)
          rd_mem_counter_disp  <= 10'b1111111101;
       else
          rd_mem_counter_disp  <= rd_mem_counter_disp + 10'd1;

    always @ (posedge clk)
       case(mem_bank_buf_buf_buf)
          1'b0:
          begin
             addr_disp_mem_0   <= rd_mem_counter_disp;
             addr_disp_mem_1   <= wr_mem_counter_disp;
          end
          1'b1:
          begin
             addr_disp_mem_0   <= wr_mem_counter_disp;
             addr_disp_mem_1   <= rd_mem_counter_disp;    //swap rd and wr;
          end
       endcase

    always @ (posedge clk) begin
       mem_bank_buf_buf_buf_buf      <= mem_bank_buf_buf_buf;
       mem_bank_buf_buf_buf_buf_buf <= mem_bank_buf_buf_buf_buf;
    end
 //logic 1;
    always @(posedge clk) begin
    //i <= mem_bank_buf_buf_buf_buf_buf;
    //if(i == 1) begin
    //d_out <= d_o_disp_mem_1;
       //end else begin
       //d_out <= d_o_disp_mem_0;
       //end
    d_out <= mem_bank_buf_buf_buf_buf_buf ? (d_o_disp_mem_1) : (d_o_disp_mem_0);
 end

 endmodule
```

Ln 513 Col 10 SystemVerilog HDL File 100% 00:00:36

BMC Files

BMC 000

```
module bmc000              // branch metric computation
(
    input     [1:0] rx_pair,
    output    [1:0] path_0_bmc,
    output    [1:0] path_1_bmc);

     wire tmp00, tmp01, tmp10, tmp11;


    assign tmp00 = rx_pair[0];
    assign tmp01 = rx_pair[1];

    assign tmp10 = ~tmp00;
    assign tmp11 = ~tmp01;

    assign path_0_bmc[0] = tmp00 ^ tmp01;
    assign path_0_bmc[1] = tmp00 & tmp01;

    assign path_1_bmc[0] = tmp10 ^ tmp11;
    assign path_1_bmc[1] = tmp10 & tmp11;
endmodule
```

BMC 001

```
module bmc001
(
    input     [1:0] rx_pair,
    output    [1:0] path_0_bmc,
    output    [1:0] path_1_bmc);


    wire tmp00, tmp01, tmp10, tmp11;


    assign tmp00 = rx_pair[0];
    assign tmp01 = ~rx_pair[1];

    assign tmp10 = ~tmp00;
    assign tmp11 = ~tmp01;

    assign path_0_bmc[0] = tmp00 ^ tmp01;
    assign path_0_bmc[1] = tmp00 & tmp01;

    assign path_1_bmc[0] = tmp10 ^ tmp11;
    assign path_1_bmc[1] = tmp10 & tmp11;


endmodule
```

BMC 010

```
module bmc010
(
    input    [1:0] rx_pair,
    output   [1:0] path_0_bmc,
    output   [1:0] path_1_bmc);

  wire tmp00, tmp01, tmp10, tmp11;


    assign tmp00 = rx_pair[0];
    assign tmp01 = ~rx_pair[1];

    assign tmp10 = ~tmp00;
    assign tmp11 = ~tmp01;

    assign path_0_bmc[0] = tmp00 ^ tmp01;
    assign path_0_bmc[1] = tmp00 & tmp01;

    assign path_1_bmc[0] = tmp10 ^ tmp11;
    assign path_1_bmc[1] = tmp10 & tmp11;
endmodule
```

BMC011

```
module bmc011
(
    input    [1:0] rx_pair,
    output   [1:0] path_0_bmc,
    output   [1:0] path_1_bmc);

  wire tmp00, tmp01, tmp10, tmp11;


    assign tmp00 = rx_pair[0];
    assign tmp01 = rx_pair[1];

    assign tmp10 = ~tmp00;
    assign tmp11 = ~tmp01;

    assign path_0_bmc[0] = tmp00 ^ tmp01;
    assign path_0_bmc[1] = tmp00 & tmp01;

    assign path_1_bmc[0] = tmp10 ^ tmp11;
    assign path_1_bmc[1] = tmp10 & tmp11;
endmodule
```

BMC100

```
module bmc100
(
    input    [1:0] rx_pair,
    output   [1:0] path_0_bmc,
    output   [1:0] path_1_bmc);


     wire tmp00, tmp01, tmp10, tmp11;


    assign tmp00 = rx_pair[0];
    assign tmp01 = rx_pair[1];

    assign tmp10 = ~tmp00;
    assign tmp11 = ~tmp01;

    assign path_0_bmc[0] = tmp00 ^ tmp01;
    assign path_0_bmc[1] = tmp00 & tmp01;

    assign path_1_bmc[0] = tmp10 ^ tmp11;
    assign path_1_bmc[1] = tmp10 & tmp11;
endmodule
```

BMC101

```
module bmc101          (
    input    [1:0] rx_pair,
    output   [1:0] path_0_bmc,
    output   [1:0] path_1_bmc);


    wire tmp00, tmp01, tmp10, tmp11;


    assign tmp00 = rx_pair[0];
    assign tmp01 = ~rx_pair[1];

    assign tmp10 = ~tmp00;
    assign tmp11 = ~tmp01;

    assign path_0_bmc[0] = tmp00 ^ tmp01;
    assign path_0_bmc[1] = tmp00 & tmp01;

    assign path_1_bmc[0] = tmp10 ^ tmp11;
    assign path_1_bmc[1] = tmp10 & tmp11;
endmodule
```

BMC110

```
module bmc110
(
    input    [1:0] rx_pair,
    output   [1:0] path_0_bmc,
    output   [1:0] path_1_bmc);


   wire tmp00, tmp01, tmp10, tmp11;


    assign tmp00 = rx_pair[0];
    assign tmp01 = ~rx_pair[1];

    assign tmp10 = ~tmp00;
    assign tmp11 = ~tmp01;

    assign path_0_bmc[0] = tmp00 ^ tmp01;
    assign path_0_bmc[1] = tmp00 & tmp01;

    assign path_1_bmc[0] = tmp10 ^ tmp11;
    assign path_1_bmc[1] = tmp10 & tmp11;
endmodule
```

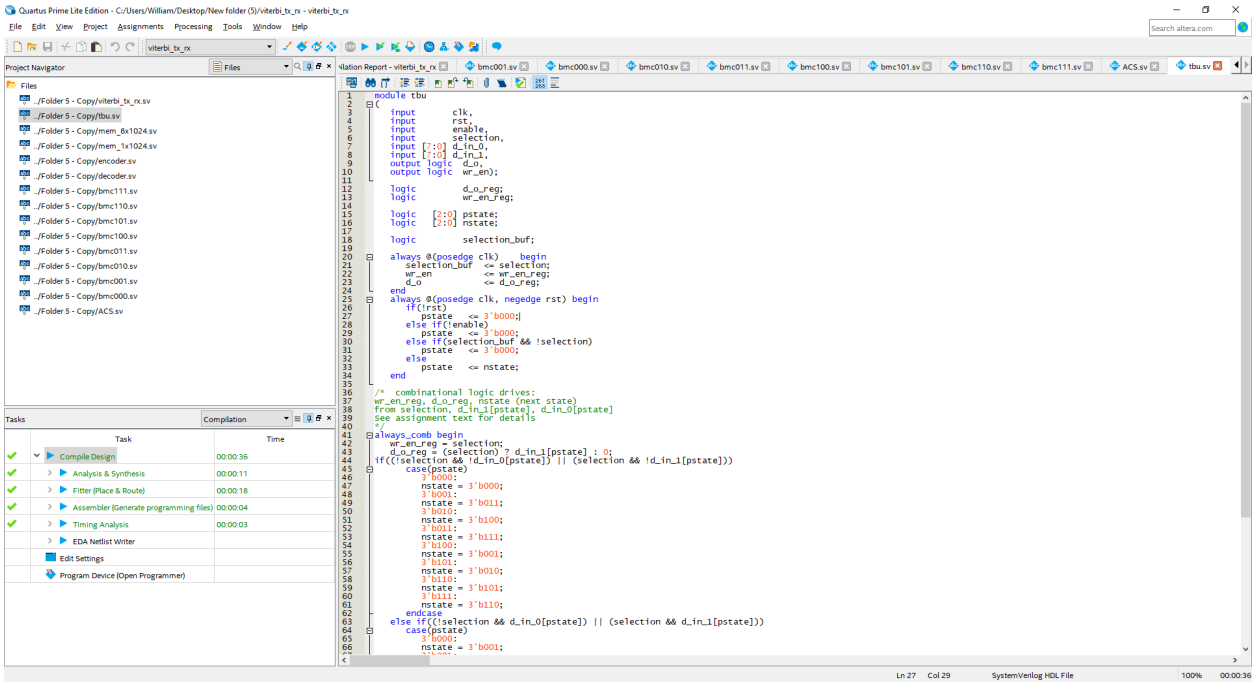
BMC111

```
module bmc111 (
    input    [1:0] rx_pair,
    output   [1:0] path_0_bmc,
    output   [1:0] path_1_bmc);


    wire tmp00, tmp01, tmp10, tmp11;


    assign tmp00 = rx_pair[0];
    assign tmp01 = rx_pair[1];

    assign tmp10 = ~tmp00;
    assign tmp11 = ~tmp01;

    assign path_0_bmc[0] = tmp00 ^ tmp01;
    assign path_0_bmc[1] = tmp00 & tmp01;

    assign path_1_bmc[0] = tmp10 ^ tmp11;
    assign path_1_bmc[1] = tmp10 & tmp11;
endmodule
```

ACS

```
module ACS                                  // add-compare-select
(
    input        path_0_valid,
    input        path_1_valid,
    input [1:0] path_0_bmc,                 // branch metric computation
    input [1:0] path_1_bmc,
    input [7:0] path_0_pmc,              // path metric computation
    input [7:0] path_1_pmc,

    output logic         selection,
    output logic         valid_o,
    output       [7:0] path_cost);

    logic [7:0] path_cost_0;                // branch metric + path metric
    logic  [7:0] path_cost_1;


    assign path_cost = (valid_o?(selection?path_cost_1:path_cost_0):8'd0);
    assign path_cost_0 = path_0_bmc + path_0_pmc;
    assign path_cost_1 = path_1_bmc + path_1_pmc;

    always_comb begin
    valid_o = (!path_0_valid && !path_1_valid) ? 1'b0 : 1'b1;

       if(path_0_valid == 0 && path_1_valid == 0) begin
          selection = 0;
          valid_o = 0;

       end

       else if(path_0_valid == 0 && path_1_valid == 1) begin
          selection = 1;
          valid_o = 1;

       end

       else if(path_0_valid == 1 && path_1_valid == 0) begin
          selection = 0;
          valid_o = 1;

       end

       else if(path_cost_0 > path_cost_1) begin
             selection = 1;
             valid_o = 1;

       end

       else begin
             selection = 0;
             valid_o = 1;

       end


    end

endmodule

```

TBU




```
else if((!selection && d_in_0[pstate]) || (selection && d_in_1[pstate]))
   case(pstate)
      3'b000:
      nstate = 3'b001;
      3'b001:
      nstate = 3'b010;
      3'b010:
      nstate = 3'b101;
      3'b011:
      nstate = 3'b110;
      3'b100:
      nstate = 3'b000;
      3'b101:
      nstate = 3'b011;
      3'b110:
      nstate = 3'b100;
      3'b111:
      nstate = 3'b111;
   endcase

      else
         nstate = pstate;
end
endmodule
```

Testbench:

Quartus Prime Lite Edition - C:/Users/William/Desktop/New folder (5)/viterbi_tx_rx - viterbi_tx_rx

File Edit View Project Assignments Processing Tools Window Help

viterbi_tx_rx

Project Navigator Hierarchy

Entity:Instanc

Arria II GX: AUTO

decoder

Tasks Compilation

Task

Compile Design

Analysis & Synthesis

Fitter (Place & Route)

Assembler (Generate programmin

Timing Analysis

EDA Netlist Writer

Edit Settings

Program Device (Open Programmer)

viterbi_tx_rx_tb(4).sv

```
// test bench
module viterbi_tx_rx_tb();
    bit clk;
    bit rst;
    bit encoder_i;          // original data
    bit enc_i_hist[2048];   // history thereof
    bit enable_encoder_i;
    wire decoder_o;         // decoded data, should match original
    bit dec_o_hist[2048];   // history thereof
    bit disp;               // end of test flag
    int good, bad;          // scoreboard

// this module contains conv. encode, channel, and Vit decode
    viterbi_tx_rx vtr(
        .clk,
        .rst,
        .encoder_i,         // original data
        .enable_encoder_i,
        .decoder_o    );    // decoded data

    always begin
        #50  clk  = 'b1;
        #50  clk  = 'b0;
    end
    int i, j, k, l;

    always @(posedge clk) begin
     enc_i_hist[i] <= encoder_i;
     i <= i+1;              // counters for data in and out
     l <= l+1;
    end

    initial begin
     #410500;//#410400;
     forever @(posedge clk) begin
       dec_o_hist[k] <= decoder_o;
       k<=k+1;
     end
    end

    initial begin
        #1000          rst          =  1'b1;
                     enable_encoder_i  =  1'b1;
        repeat(2) begin
        #100          encoder_i=  1'b1;
        #100          encoder_i=  1'b0;
        #200          encoder_i=  1'b1;
        #200          encoder_i=  1'b0;
        #300          encoder_i=  1'b1;
        #300          encoder_i=  1'b0;
        #400          encoder_i=  1'b1;
        #400          encoder_i=  1'b0;
        #500         encoder_i=  1'b1;
        #500          encoder_i=  1'b0;
        #100          encoder_i=  1'b1;
        #100          encoder_i=  1'b0;
        #100          encoder_i=  1'b1;
        #100          encoder_i=  1'b0;
        #100          encoder_i=  1'b1;
        #100          encoder_i=  1'b0;
        #100          encoder_i=  1'b1;
        #100          encoder_i=  1'b0;
        #100          encoder_i=  1'b1;
        #100          encoder_i=  1'b0;
        end
        #1000       encoder_i=  1'b1;
        #1000       encoder_i=  1'b0;
        repeat(20)
```

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Quartus Prime Lite Edition - C:/Users/William/Desktop/New folder (5)/viterbi_tx_rx - viterbi_tx_rx

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viterbi_tx_rx

Project Navigator Hierarchy

Entity:Instanc

Arria II GX: AUTO

decoder

Tasks Compilation

Task

Compile Design

Analysis & Synthesis

Fitter (Place & Route)

Assembler (Generate programmin

Timing Analysis

EDA Netlist Writer

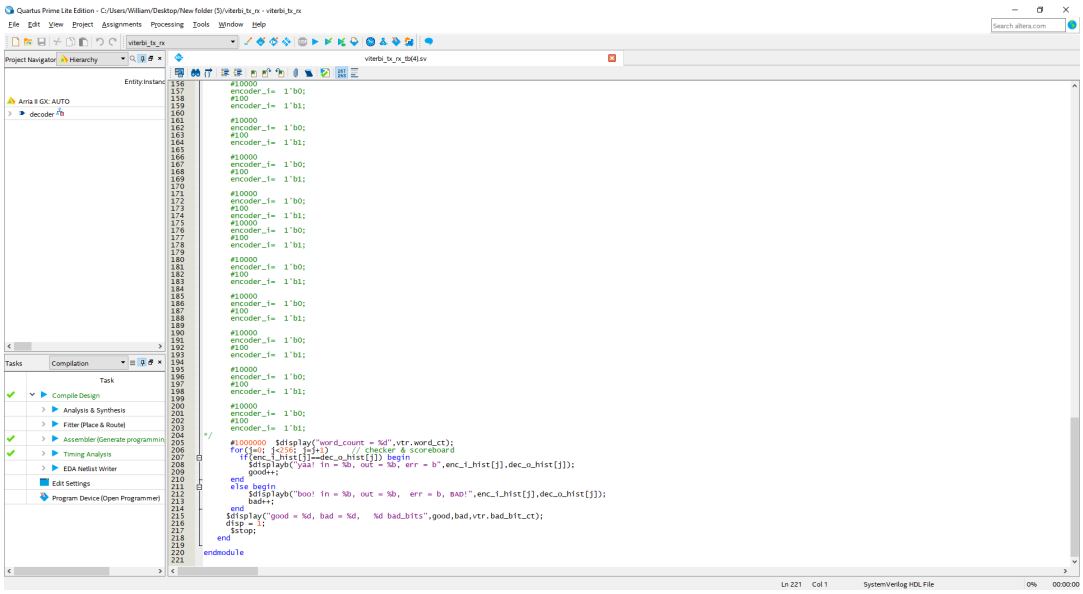
Edit Settings

Program Device (Open Programmer)

viterbi_tx_rx_tb(4).sv

```
        repeat(20)
        #100          encoder_i=  $random>>1;
        #100          encoder_i=  1'b0;
        #1000          encoder_i=  1'b1;
        #1000         encoder_i=  1'b0;
        #100          encoder_i=  1'b1;
        #10000         encoder_i=  1'b0;
        #100           encoder_i=  1'b1;
        #10000         encoder_i=  1'b0;
        #100           encoder_i=  1'b1;
        #1000         encoder_i=  1'b0;
/*          #100          encoder_i=  1'b1;
        #10000         encoder_i=  1'b0;
        #100           encoder_i=  1'b1;
        #10000         encoder_i=  1'b0;
        #100           encoder_i=  1'b1;
        #10000         encoder_i=  1'b0;
        #100           encoder_i=  1'b1;
        #10000         encoder_i=  1'b0;
        #100          encoder_i=  1'b1;
        #10000        encoder_i=  1'b0;
        #100          encoder_i=  1'b1;
        #10000         encoder_i=  1'b0;
        #100           encoder_i=  1'b1;
        #10000         encoder_i=  1'b0;
        #100           encoder_i=  1'b1;

        #10000
        encoder_i=  1'b0;
        #100
        encoder_i=  1'b1;

        #10000
        encoder_i=  1'b0;
        #100
        encoder_i=  1'b1;

        #10000
        encoder_i=  1'b0;
        #100
        encoder_i=  1'b1;

        #10000
        encoder_i=  1'b0;
        #100
        encoder_i=  1'b1;

        #10000
        encoder_i=  1'b0;
        #100
        encoder_i=  1'b1;

        #10000
        encoder_i=  1'b0;
        #100
        encoder_i=  1'b1;

        #10000   encoder_i=  1'b0;
```

Ln 112 Col 24 SystemVerilog HDL File 0% 00:00:00

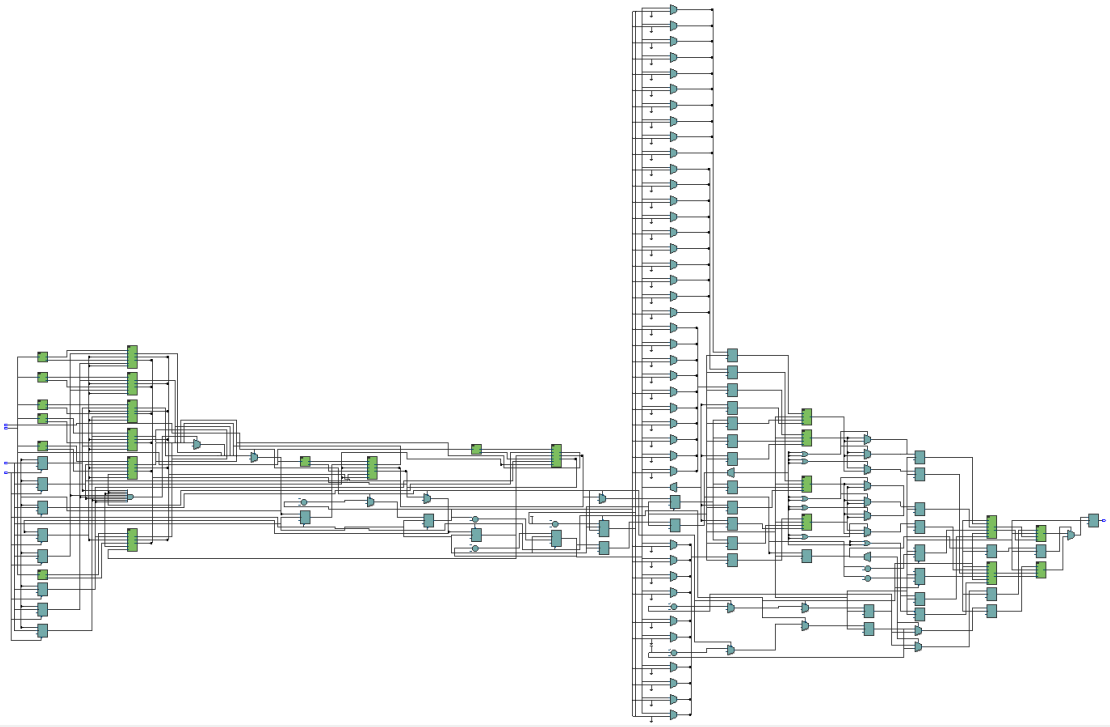


Simulation Transcript

```
# yaa! in = 1, out = 1, err = b
# good =          256, bad =            0,              0 bad_bits
# ** Note: $stop    : C:/Users/William/Desktop/New folder (7)/viterbi_tx_rx_tb(4).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at C:/Users/William/Desktop/New folder (7)/viterbi_tx_rx_tb(4).sv line 217

VSIM 3>
```

Proof of synthesis

RTL Viewer



Transcript Log

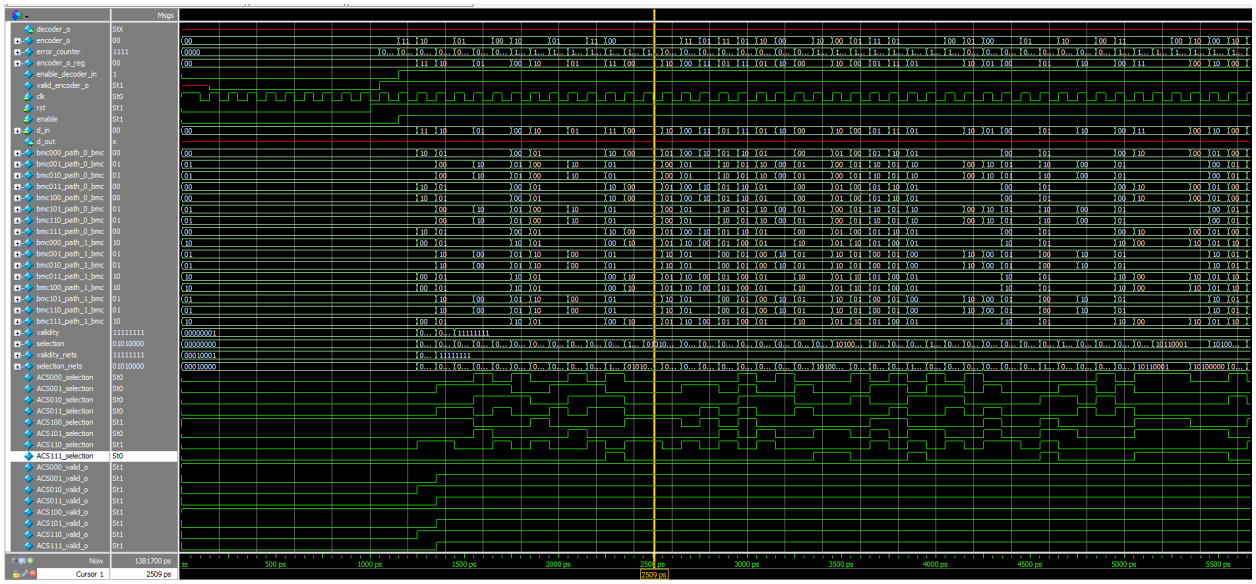
```
# Reading pref.tcl
# Loading project Part 2
# Compile of ACS.sv was successful.
# Compile of bmc000.sv was successful.
# Compile of bmc001.sv was successful.
# Compile of bmc010.sv was successful.
# Compile of bmc011.sv was successful.
# Compile of bmc100.sv was successful.
# Compile of bmc101.sv was successful.
# Compile of bmc110.sv was successful.
# Compile of bmc111.sv was successful.
# Compile of decoder.sv was successful.
# Compile of encoder.sv was successful.
# Compile of mem_1x1024.sv was successful.
# Compile of mem_8x1024.sv was successful.
# Compile of tbu.sv was successful.
# Compile of viterbi_tx_rx_tb(4).sv was successful.
# Compile of viterbi_tx_rx1(1).sv was successful.
# 16 compiles, 0 failed with no errors.
# Compile of ACS.sv was successful.
# Compile of bmc000.sv was successful.
# Compile of bmc001.sv was successful.
# Compile of bmc010.sv was successful.
# Compile of bmc011.sv was successful.
# Compile of bmc100.sv was successful.
# Compile of bmc101.sv was successful.
# Compile of bmc110.sv was successful.
# Compile of bmc111.sv was successful.
# Compile of decoder.sv was successful.
# Compile of encoder.sv was successful.
# Compile of mem_1x1024.sv was successful.
# Compile of mem_8x1024.sv was successful.
# Compile of tbu.sv was successful.
# Compile of viterbi_tx_rx_tb(4).sv was successful.
# Compile of viterbi_tx_rx1(1).sv was successful.
# 16 compiles, 0 failed with no errors.
ModelSim> vsim -gui work.viterbi_tx_rx_tb
# vsim -gui work.viterbi_tx_rx_tb
# Start time: 16:18:31 on Mar 22,2024
# Loading sv_std.std
# Loading work.viterbi_tx_rx_tb
# Loading work.viterbi_tx_rx
# Loading work.encoder
# Loading work.decoder
# Loading work.bmc000
# Loading work.bmc001
# Loading work.bmc010
# Loading work.bmc011
# Loading work.bmc100
# Loading work.bmc101
# Loading work.bmc110
# Loading work.bmc111
# Loading work.ACS
# Loading work.mem
# Loading work.tbu
# Loading work.mem_disp
# WARNING: No extended dataflow license exists
VSIM 2> run -all
# word count =        10440
```

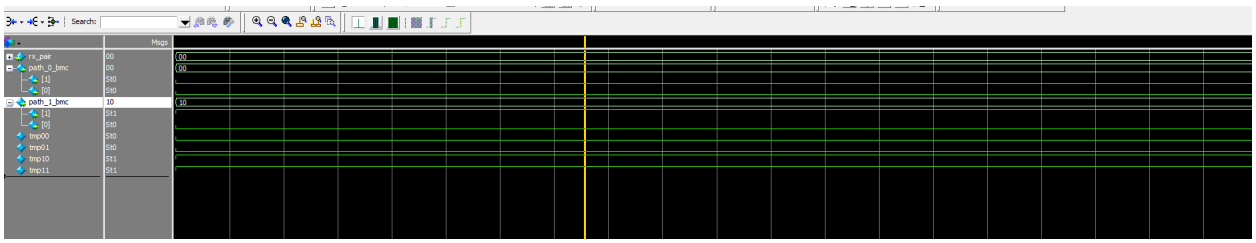
Encoder explanation

In the term project encoder file, the encoded output, d_out, is determined by the current input, d_in. The current state, cstate, and the next state are 3 bits wide and the next state is dependent on the d_in. In the homework 7 file, there is a different structure where it is a little more flexible. With the use of two different mask patterns, mask0 and mask1, and the inclusion of the shift register, the output will produce more encrypted codes based on the loaded mask. Unlike the encoder file, the data_out will be controlled by the two mask patterns instead of just the input. The main reason why the decoder does not work with the homework 7 encoder is because the Viterbi decoder is created to work with specific convolutional code. As we mentioned before, the homework 7 encoder can create a variety of convolutional codes and that makes the decoder unable to process it. Whereas, the encoder.sv file is a fixed encoder which is tailored to this specific decoder. The state transition logic is matched and for the decoder to process. To elaborate further on these codes, the homework file is also non-systematic and non-recursive because there is no feedback output in the shift registers. Additionally, the shift register does not copy the input bits to the output bits that come from the XOR operation and position the shift register uniquely. The encoder file is systematic and recursive because it has feedback and due to d_out_reg[0] = d_in proves that there is functionality behind that. This is important because systematic codes are easier for decoders to handle because the input data has a correlation with the original data whereas non systematic codes have no direct reference to the actual data bit.

Decoder explanation

The decoder receives the encoded data from the encoder and will begin the path metric calculation within the BMC that determines the path cost of each state. Next the ACS, Add-Compare-Select, will add each possible transition into the states and compare the metrics of the two paths of which one has a higher chance of a lower cost which is dependent on the amount of error. The ACS also selects the better path, lower path cost, and sets the path as a more cost effective path. In the trellis memory module, it represents the survivor path selection that will determine the most cost effective path leading to each state. Throughout this process, the decoder eliminates other less likely paths and continuously updates the path metrics which will prepare the best route for the traceback unit. In the traceback unit, the sequence should be entirely processed in which the decoder should find the best ending state with the least path cost. The Trellis module should already have highlighted the most likely path and reconstructed the original data sequence. Finally, the d_out should output the most likely path of the unencoded data bit.





When no errors present, the minimum branch metric = 0 due to the hamming distance of the input. In BMC, the input, rx_pair, is set to the temp variables and assigned to either path 0 or path 1 which is then ANDed and XORed. You can see that if the BMC output on a path was 00 that meant most likely that it had the least path cost which would point out to the decoder that it would be the most likely path.

There are various steps on how the decoder determines the sequence of bits that was sent from the encoder. The branch metric, BMC, computes the transition between the states at each clock cycle. Each BMC will measure the difference between the received encoder output, d_in. In ACS, it updates the path's metric cost and points out which is more optimal. The ACS is able to point out the optimal path by comparing the path_cost_0 or path_cost_1 and selection will choose one or the other by assigning itself to 1 or 0. Once the entire sequence has been processed the traceback unit tries to reconstruct the original sequence from each nstate by setting a temp variable pstate to either the nstate or 0 when reset is 0, enable is 0, or when selection_buf && !selection passes through. This traceback unit is hard coded as depending on the whether the selection and pstate, it outputs a specific nstate. The nstate value will be outputted dependent on the memory module operation that will decide which nstate path is correct.

Part 2: Injecting the bad bits

Disclaimer: We used a different testbench from the one that was provided because we worked on it earlier. All data and results were confirmed with the TA and it worked just as expected.

Testbench:

```
// test bench
module viterbi_tx_rx_tb();
   bit clk;
   bit rst;
   bit encoder_i;        // original data
   bit enc_i_hist[2048];   // history thereof
   bit enable_encoder_i;
   wire decoder_o;         // decoded data
   bit dec_o_hist[2048];   // history thereof
   bit disp;
   int good, bad;          // scoreboard

// this module contains conv. encode and Vit decode
   viterbi_tx_rx vtr(
      .clk,
      .rst,
      .encoder_i,        // original data
      .enable_encoder_i,
      .decoder_o    );     // decoded data

   always begin
      #50   clk    = 1;
      #50  clk     = 0;
   end
   int i, j, k, l;

   always @(posedge clk) begin
    enc_i_hist[i] <= encoder_i;
    i <= i+1;
    l <= l+1;
   end
      initial begin
  #410500;//#410400;
  forever @(posedge clk) begin
    dec_o_hist[k] <= decoder_o;
    k<=k+1;
     end
   end

   initial begin
      clk       =  1'b1;
      #1000
      rst        =  1'b1;
      enable_encoder_i  =  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
```

```
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;

      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b1;
      #100
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b0;
```

```
131         encoder_i=  1'b0;
132         #100
133         encoder_i=  1'b0;
134         #100
135         encoder_i=  1'b0;
136         #100
137         encoder_i=  1'b1;
138         #100
139         encoder_i=  1'b1;
140         #100
141         encoder_i=  1'b1;
142         #100
143         encoder_i=  1'b1;
144         #100
145         encoder_i=  1'b0;
146         #100
147         encoder_i=  1'b0;
148         #100
149         encoder_i=  1'b0;
150         #100
151         encoder_i=  1'b0;
152         #100
153         encoder_i=  1'b0;
154         #100
155         encoder_i=  1'b1;
156         #100
157         encoder_i=  1'b1;
158         #100
159         encoder_i=  1'b1;
160         #100
161         encoder_i=  1'b1;
162         #100
163         encoder_i=  1'b1;
164
165         #10000
166         encoder_i=  1'b0;
167         #100
168         encoder_i=  1'b1;
169         #1000
170         encoder_i=  1'b0;
171         #100
172         encoder_i=  1'b1;
173
174         #10000
175         encoder_i=  1'b0;
176         #100
177         encoder_i=  1'b1;
178
179
180         #10000
181         encoder_i=  1'b0;
182         #100
183         encoder_i=  1'b1;
184
185         #10000
186         encoder_i=  1'b0;
187         #100
188         encoder_i=  1'b1;
189
190         #10000
191         encoder_i=  1'b0;
192         #100
193         encoder_i=  1'b1;
194
195         #10000
196         encoder_i=  1'b0;
```

```
196         encoder_i=  1'b0;
197         #100
198         encoder_i=  1'b1;
199
200
201         #10000
202         encoder_i=  1'b0;
203         #100
204         encoder_i=  1'b1;
205
206         #10000
207         encoder_i=  1'b0;
208         #100
209         encoder_i=  1'b1;
210
211         #10000
212         encoder_i=  1'b0;
213         #100
214         encoder_i=  1'b1;
215
216         #10000
217         encoder_i=  1'b0;
218         #100
219         encoder_i=  1'b1;
220
221
222         #10000
223         encoder_i=  1'b0;
224         #100
225         encoder_i=  1'b1;
226
227         #10000
228         encoder_i=  1'b0;
229         #100
230         encoder_i=  1'b1;
231
232         #10000
233         encoder_i=  1'b0;
234         #100
235         encoder_i=  1'b1;
236
237         #10000
238         encoder_i=  1'b0;
239         #100
240         encoder_i=  1'b1;
241
242         #10000
243         encoder_i=  1'b0;
244         #100
245         encoder_i=  1'b1;
246
247
248         #10000
249         encoder_i=  1'b0;
250         #100
251         encoder_i=  1'b1;
252
253         #10000
254         encoder_i=  1'b0;
255         #100
256         encoder_i=  1'b1;
257
258         #10000
259         encoder_i=  1'b0;
260         #100
261         encoder_i=  1'b1;
```

```
261         encoder_i=  1'b1;
262
263         #10000
264         encoder_i=  1'b0;
265         #100
266         encoder_i=  1'b1;
267
268         #10000
269         encoder_i=  1'b0;
270         #100
271         encoder_i=  1'b1;
272
273         #10000
274         encoder_i=  1'b0;
275         #100
276         encoder_i=  1'b1;
277
278         #10000
279         encoder_i=  1'b0;
280         #100
281         encoder_i=  1'b1;
282
283         #10000
284         encoder_i=  1'b0;
285         #100
286         encoder_i=  1'b1;
287
288         #10000
289         encoder_i=  1'b0;
290         #100
291         encoder_i=  1'b1;
292
293         #10000
294         encoder_i=  1'b0;
295         #100
296         encoder_i=  1'b1;
297
298
299         #10000
300         encoder_i=  1'b0;
301         #100
302         encoder_i=  1'b1;
303
304         #10000
305         encoder_i=  1'b0;
306         #100
307         encoder_i=  1'b1;
308
309         #10000
310         encoder_i=  1'b0;
311         #100
312         encoder_i=  1'b1;
313
314         #10000
315         encoder_i=  1'b0;
316         #100
317         encoder_i=  1'b1;
318
319         #10000
320         encoder_i=  1'b0;
321         #100
322         encoder_i=  1'b1;
323
324         #10000
325         encoder_i=  1'b0;
326         #100
```

```
326         #100
327         encoder_i=  1'b1;
328         #10000
329         encoder_i=  1'b0;
330         #100
331         encoder_i=  1'b1;
332
333         #10000
334         encoder_i=  1'b0;
335         #100
336         encoder_i=  1'b1;
337
338         #10000
339         encoder_i=  1'b0;
340         #100
341         encoder_i=  1'b1;
342
343         #10000
344         encoder_i=  1'b0;
345         #100
346         encoder_i=  1'b1;
347
348         #10000
349         encoder_i=  1'b0;
350         #100
351         encoder_i=  1'b1;
352
353         #10000
354         encoder_i=  1'b0;
355         #100
356         encoder_i=  1'b1;
357
358         #1000000
359         for(j=0; j<256; j=j+1)      // checker & scoreboard
360           if(enc_i_hist[j]==dec_o_hist[j]) begin
361             $displayb("yaa! in = %b, out = %b",enc_i_hist[j],dec_o_hist[j]);
362             good++;
363         end
364         else begin
365             $displayb("boo! in = %b, out = %b",enc_i_hist[j],dec_o_hist[j]);
366             bad++;
367         end
368       $display("good = %d, bad = %d",good,bad);
369       disp = 1;
370
371         $stop;
372     end
373
374   endmodule
375
```

Part A

2.a.1

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic        enable_decoder_in;
    wire         valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 1;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
           error_counter  <= 4'd0;
           encoder_o_reg  <= 2'b00;
           enable_decoder_in <= 1'b0;
           error_bit8 <= 3'd0;
           error_bit32 <= 5'd0;
            word_ct = 0;
       end
       else     begin
           enable_decoder_in <= valid_encoder_o;
           encoder_o_reg  <= 2'b00;
           error_counter  <= error_counter + 4'd1;
           error_bit8 <= error_bit8 + 3'd1;
           error_bit32 <= error_bit32 + 5'd1;
            word_ct              <= word_ct + 1;

         if(error_bit8==3'b111)begin //2a1 # good =         256, bad =           0 , Injected : 32
             encoder_o_reg  <= {encoder_o[1],~encoder_o[0]};  // inject one bad bit out of every 8
                if(word_ct<256)begin
                bad_bit_ct      <= bad_bit_ct + 1;
                $display("error_counter = %d",bad_bit_ct);
               end
           end

           else
               encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

  // insert your convolutional encoder here
  // change port names and module name as necessary/desired
    encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );
```

```
# error_counter =            30
# error_counter =            31
# error_counter =            32
```

```
# good =          256, bad =            0
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
```

```
       .d_out(encoder_o)   );

  // insert your term project code here
    decoder decoder1       (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

  endmodule
```

2.a.2

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic       enable_decoder_in;
    wire        valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 1;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter  <= 4'd0;
          encoder_o_reg  <= 2'b00;
          enable_decoder_in <= 1'b0;
          error_bit8 <= 3'd0;
          error_bit32 <= 5'd0;
            word_ct = 0;
       end
       else     begin
          enable_decoder_in <= valid_encoder_o;
          encoder_o_reg  <= 2'b00;
          error_counter  <= error_counter + 4'd1;
          error_bit8 <= error_bit8 + 3'd1;
          error_bit32 <= error_bit32 + 5'd1;
            word_ct              <= word_ct + 1;

          if(error_bit8==3'b111)begin //2a2 # good =         256, bad =           0 , Injected : 32
             encoder_o_reg  <= {~encoder_o[1],encoder_o[0]};  // inject one bad bit out of every 8
               if(word_ct<256)begin
                 bad_bit_ct      <= bad_bit_ct + 1;
                 $display("error_counter = %d",bad_bit_ct);
              end
          end

          else
             encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

  // insert your convolutional encoder here
  // change port names and module name as necessary/desired
    encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );
```

```
       end

  // insert your convolutional encoder here
  // change port names and module name as necessary/desired
    encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );

  // insert your term project code here
    decoder decoder1       (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

 endmodule
```

```
# error_counter =            30
# error_counter =            31
# error_counter =            32
```

```
# yaa! in = 1, out = 1
# good =          256, bad =            0
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv line 371
```

2.a.3

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic       enable_decoder_in;
    wire        valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 2;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter  <= 4'd0;
          encoder_o_reg  <= 2'b00;
          enable_decoder_in <= 1'b0;
          error_bit8 <= 3'd0;
          error_bit32 <= 5'd0;
           word_ct = 0;
       end
       else     begin
          enable_decoder_in <= valid_encoder_o;
          encoder_o_reg  <= 2'b00;
          error_counter  <= error_counter + 4'd1;
          error_bit8 <= error_bit8 + 3'd1;
          error_bit32 <= error_bit32 + 5'd1;
           word_ct                <= word_ct + 1;

          if(error_bit8==3'b111)begin //2a3 # good =          205, bad =          51 , Injected = 64
             encoder_o_reg  <= {~encoder_o[1],~encoder_o[0]};    // inject one bad bit for 1 and 0 out of every 8
                if(word_ct<256)begin
                   bad_bit_ct     <= bad_bit_ct + 2;
                      $display("error_counter = %d",bad_bit_ct);
                end
          end

          else
             encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

  // insert your convolutional encoder here
  // change port names and module name as necessary/desired
    encoder encoder1          (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );

  // insert your term project code here
    decoder decoder1        (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

  endmodule
```

```
# error_counter =          58
# error_counter =          60
# error_counter =          62
# error_counter =          64
# yaa! in = 0, out = 0
# yaa! in = 0, out = 0
```

```
# boo! in = 1, out = 0
# good =          205, bad =          51
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv line 371
```

2.a.4

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic       enable_decoder_in;
    wire        valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 1;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
           error_counter   <= 4'd0;
           encoder_o_reg  <= 2'b00;
           enable_decoder_in <= 1'b0;
           error_bit8 <= 3'd0;
           error_bit32 <= 5'd0;
             word_ct = 0;
       end
       else     begin
           enable_decoder_in <= valid_encoder_o;
           encoder_o_reg  <= 2'b00;
           error_counter  <= error_counter + 4'd1;
           error_bit8 <= error_bit8 + 3'd1;
           error_bit32 <= error_bit32 + 5'd1;
             word_ct              <= word_ct + 1;

         if(error_counter == 4'b1001 || error_counter == 4'b1000 ) begin  //2a4 # good =     256, bad =     0 , Injected : 32
             encoder_o_reg  <= {encoder_o[1],~encoder_o[0]};  // inject two bad bit out of every 16
                 if(word_ct<256)begin
                     bad_bit_ct      <= bad_bit_ct + 1;
                     $display("error_counter = %d",bad_bit_ct);
              end
         end

          else
             encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

  // insert your convolutional encoder here
  // change port names and module name as necessary/desired
    encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );
```

```
# error_counter =          29
# error_counter =          30
# error_counter =          31
# error_counter =          32
# yaa! in = 0, out = 0
# yaa! in = 0, out = 0
# yaa! in = 1, out = 1
# good =          256, bad =            0
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv line 371
```

2.a.5

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic       enable_decoder_in;
    wire        valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 1;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
           error_counter  <= 4'd0;
           encoder_o_reg  <= 2'b00;
           enable_decoder_in <= 1'b0;
           error_bit8 <= 3'd0;
           error_bit32 <= 5'd0;
            word_ct = 0;
       end
       else     begin
           enable_decoder_in <= valid_encoder_o;
           encoder_o_reg  <= 2'b00;
           error_counter  <= error_counter + 4'd1;
           error_bit8 <= error_bit8 + 3'd1;
           error_bit32 <= error_bit32 + 5'd1;
            word_ct                 <= word_ct + 1;

             if(error_counter == 4'b1001 || error_counter == 4'b1000 ) begin  //2a5 # good =     256, bad =      0 , Injected : 32
              encoder_o_reg  <= {~encoder_o[1],encoder_o[0]};  // inject two bad bit out of every 16
                 if(word_ct<256)begin
                    bad_bit_ct      <= bad_bit_ct + 1;
                    $display("error_counter = %d",bad_bit_ct);
              end
         end

          else
              encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

 // insert your convolutional encoder here
 // change port names and module name as necessary/desired
   encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );

 // insert your term project code here
   decoder decoder1       (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

 endmodule
```

```
# error_counter =          29
# error_counter =          30
# error_counter =          31
# error_counter =          32
# yaa! in = 0, out = 0
# yaa! in = 0, out = 0
# yaa! in = 1, out = 1
# good =         256, bad =           0
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv line 371
```

2.a.6

```sv
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic        enable_decoder_in;
    wire         valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 1;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
           error_counter   <= 4'd0;
           encoder_o_reg  <= 2'b00;
           enable_decoder_in <= 1'b0;
           error_bit8 <= 3'd0;
           error_bit32 <= 5'd0;
             word_ct = 0;
       end
       else      begin
           enable_decoder_in <= valid_encoder_o;
           encoder_o_reg  <= 2'b00;
           error_counter  <= error_counter + 4'd1;
           error_bit8 <= error_bit8 + 3'd1;
           error_bit32 <= error_bit32 + 5'd1;
            word_ct                <= word_ct + 1;

            if(error_bit32 == 5'b01010 || error_bit32 == 5'b01011 || error_bit32 == 5'b01100 || error_bit32 == 5'b01101 )begin //2a6 //# good = 223, bad =  33 Injected :32
              encoder_o_reg  <= {encoder_o[1],~encoder_o[0]};  // inject 4 in a row bad bit out of every 32
                  if(word_ct<256)begin
                    bad_bit_ct      <= bad_bit_ct + 1;
                    $display("error_counter = %d",bad_bit_ct);
                   end
               end
           else
               encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

  // insert your convolutional encoder here
  // change port names and module name as necessary/desired
    encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );


  // insert your term project code here
    decoder decoder1       (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

  endmodule
```

```
# error_counter =          28
# error_counter =          29
# error_counter =          30
# error_counter =          31
# error_counter =          32
# yaa! in = 0, out = 0
# yaa! in = 1, out = 1
# good =         223, bad =          33
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv line 371
```

2.a.7

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic       enable_decoder_in;
    wire        valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 1;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter  <= 4'd0;
          encoder_o_reg  <= 2'b00;
          enable_decoder_in <= 1'b0;
          error_bit8 <= 3'd0;
          error_bit32 <= 5'd0;
           word_ct = 0;
       end
       else     begin
          enable_decoder_in <= valid_encoder_o;
          encoder_o_reg  <= 2'b00;
          error_counter  <= error_counter + 4'd1;
          error_bit8 <= error_bit8 + 3'd1;
          error_bit32 <= error_bit32 + 5'd1;
           word_ct              <= word_ct + 1;

          if(error_bit32 == 5'b01010 || error_bit32 == 5'b01011 || error_bit32 == 5'b01100 || error_bit32 == 5'b01101 )begin //2a7 // # good = 240, bad =  16, Injected :32
             encoder_o_reg  <= {~encoder_o[1],encoder_o[0]};  // inject 4 in a row bad bit out of every 32
              if(word_ct<256)begin
                bad_bit_ct     <= bad_bit_ct + 1;
                $display("error_counter = %d",bad_bit_ct);
              end
          end

          else
             encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

  // insert your convolutional encoder here
  // change port names and module name as necessary/desired
  encoder encoder1        (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );
```

```
  // insert your term project code here
  decoder decoder1        (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

endmodule
```

```
# error_counter =          31
# error_counter =          32
# good =          240, bad =          16
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv line 371
```

2.a.8

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic        enable_decoder_in;
    wire         valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 2;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter  <= 4'd0;
          encoder_o_reg  <= 2'b00;
          enable_decoder_in <= 1'b0;
          error_bit8 <= 3'd0;
          error_bit32 <= 5'd0;
           word_ct = 0;
       end
       else      begin
          enable_decoder_in <= valid_encoder_o;
          encoder_o_reg  <= 2'b00;
          error_counter  <= error_counter + 4'd1;
          error_bit8 <= error_bit8 + 3'd1;
          error_bit32 <= error_bit32 + 5'd1;
           word_ct                <= word_ct + 1;

           if(error_bit32 == 5'b01010 || error_bit32 == 5'b01011  ) begin //2a8 # good =         232, bad =          24 Injected: 32
              encoder_o_reg  <= {~encoder_o[1],~encoder_o[0]};    // inject 2 in a row bad bit out of every 32
                  if(word_ct<256)begin
                    bad_bit_ct      <= bad_bit_ct + 2;
                     $display("error_counter = %d",bad_bit_ct);
                   end
             end

          else
              encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

 // insert your convolutional encoder here
 // change port names and module name as necessary/desired
    encoder encoder1        (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );

 // insert your term project code here
    decoder decoder1        (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

 endmodule
```

```
# error_counter =           28
# error_counter =           30
# error_counter =           32
# yaa! in = 0, out = 0
```

```
# good =          232, bad =           24
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb.sv line 371
```

Part B

Testbench

```
// test bench
module viterbi_tx_rx_tb();
   bit clk;
   bit rst;
   bit encoder_i;          // original data
   bit enc_i_hist[2048];   // history thereof
   bit enable_encoder_i;
   wire decoder_o;         // decoded data, should match original
   bit dec_o_hist[2048];   // history thereof
   bit disp;             // end of test flag
   int good, bad;          // scoreboard

 // this module contains conv. encode, channel, and vit decode
   viterbi_tx_rx vtr(
      .clk,
      .rst,
      .encoder_i,         // original data
      .enable_encoder_i,
      .decoder_o    );    // decoded data

   always begin
      #50  clk   = 'b1;
      #50  clk   = 'b0;
   end
   int i, j, k, l;

   always @(posedge clk) begin
    enc_i_hist[i] <= encoder_i;
    i <= i+1;             // counters for data in and out
    l <= l+1;
   end

   initial begin
    #410500;//#410400;
    forever @(posedge clk) begin
      dec_o_hist[k] <= decoder_o;
      k<=k+1;
     end
   end

   initial begin
      #1000       rst        =  1'b1;
                  enable_encoder_i   =  1'b1;
      repeat(2) begin
      #100        encoder_i=  1'b1;
      #100        encoder_i=  1'b0;
      #200        encoder_i=  1'b1;
      #200        encoder_i=  1'b0;
      #300        encoder_i=  1'b1;
      #300        encoder_i=  1'b0;
      #400        encoder_i=  1'b1;
      #400        encoder_i=  1'b0;
      #500       encoder_i=  1'b1;
      #500        encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #100        encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #100        encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #100        encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #100        encoder_i=  1'b0;
      end
      #1000      encoder_i=  1'b1;
      #1000      encoder_i=  1'b0;
     repeat(20)
```

```
     repeat(20)
      #100       encoder_i=  $random>>3;
      #100       encoder_i=  1'b0;
      #1000       encoder_i=  1'b1;
      #1000      encoder_i=  1'b0;
      #100       encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
/*        #100       encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
      #100       encoder_i=  1'b1;
      #10000     encoder_i=  1'b0;
      #100       encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
      #100        encoder_i=  1'b1;
      #10000      encoder_i=  1'b0;
      #100        encoder_i=  1'b1;

      #10000
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;
      #10000
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;

      #10000
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;

      #10000
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;

      #10000
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;

      #10000
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;

      #10000
      encoder_i=  1'b0;
      #100
      encoder_i=  1'b1;

      #10000   encoder_i=  1'b0;
```

```
      encoder_i=  1'b1;
*/
      #1000000  $display("word_count = %d",vtr.word_ct);
      for(j=0; j<256; j=j+1)      // checker & scoreboard
        if(enc_i_hist[j]==dec_o_hist[j]) begin
          $displayb("yaa! in = %b, out = %b, err = %b",enc_i_hist[j],dec_o_hist[j],vtr.err_inj);
          good++;
      end
      else begin
          $displayb("boo! in = %b, out = %b,  err = %b, BAD!",enc_i_hist[j],dec_o_hist[j],vtr.err_inj);
          bad++;
      end
     $display("good = %d, bad = %d,   %d bad_bits",good,bad,vtr.bad_bit_ct);
     disp = 1;
      $stop;
   end

endmodule
```

B initial Parameter changes

Varun told us to add this

Injecting 2 bad bits at [1] and [0] for N = 1

```
# good =          187, bad =           69,            108 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

Injecting 2 bad bits at [1] and [0] for N = 2

```
# good =          246, bad =           10,             41 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

Injecting 2 bad bits at [1] and [0] for N = 3

```
# good =          252, bad =            4,             14 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

Injecting 2 bad bits at [1] and [0] for N = 4

```
# good =          256, bad =            0,              5 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

2b1.

```
// contains convolutional encoder,
//  (possibly corrupting) channel,
//  and viterbi decoder
// parameter N sets the channel bit error rate
// this time, two errors in a row
module viterbi_tx_rx #(parameter N=3) (
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;  // connects encoder to decoder

    int           error_counter,
                  error_counterQ,
                  bad_bit_ct,
                  word_ct;
    logic   [1:0] encoder_o_regQ,
                  encoder_o_reg;
    logic         encoder_i_reg;
    logic         enable_decoder_in;
    logic         enable_encoder_i_reg;
    wire          valid_encoder_o;
    logic   [1:0] err_inj;

    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter        <= 'd0;
//      error_counterQ       <= 'd0;
          encoder_o_reg        <= 'b0;
        encoder_o_regQ       <= 'b0;
          enable_decoder_in    <= 'b0;
        enable_encoder_i_reg <= 'b0;
        word_ct              <= 'b0;
       end
       else begin
          enable_encoder_i_reg <= enable_encoder_i;
          enable_decoder_in    <= valid_encoder_o;
//          encoder_o_reg        <= 'b0;
          if(word_ct[0]) error_counter        <= $random;
          word_ct              <= word_ct + 1;
// bit error injection in encoder_o_reg
          encoder_i_reg     <= encoder_i;
          encoder_o_regQ    <= encoder_o;
//      error_counterQ    <= error_counter;
          if(error_counter[N-1:0]== 1) begin //
           err_inj[0]       <= error_counter[28];
             encoder_o_reg  <= encoder_o^err_inj;    // inject bad bits
          end
          else begin                  // clean version
             encoder_o_reg  <= encoder_o;
             err_inj        <= 2'b0;
       end
         if(word_ct<256) begin
           bad_bit_ct  <= bad_bit_ct + (encoder_o_regQ[1]^encoder_o_reg[1])
                                     + (encoder_o_regQ[0]^encoder_o_reg[0]);
         $display("error_counter,err_inj = %h %b %d %d",
                 error_counter,err_inj,bad_bit_ct,word_ct);
         end
       end

// insert your convolutional encoder here
// change port names and module name as necessary/desired
  encoder encoder1       (
      .clk,
      .rst,
      .enable_i(enable_encoder_i), //_reg),
      .d_in     (encoder_i),       //_reg),
      .valid_o (valid_encoder_o),
      .d_out   (encoder_o)    );

// insert your term project code here
  decoder decoder1       (
      .clk,
      .rst,
      .enable (enable_decoder_in),
      .d_in   (encoder_o_reg),
      .d_out  (decoder_o)   );

endmodule
```

```
# good =          256, bad =            0,              6 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

2b2.

```
// contains convolutional encoder,
//  (possibly corrupting) channel,
//  and viterbi decoder
// parameter N sets the channel bit error rate
// this time, two errors in a row
module viterbi_tx_rx #(parameter N=3) (
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;  // connects encoder to decoder

    int           error_counter,
                  error_counterQ,
                  bad_bit_ct,
                  word_ct;
    logic   [1:0] encoder_o_reg0,
                  encoder_o_reg;
    logic         encoder_i_reg;
    logic         enable_decoder_in;
    logic         enable_encoder_i_reg;
    wire          valid_encoder_o;
    logic   [1:0] err_inj;

    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter        <= 'd0;
//      error_counterQ       <= 'd0;
          encoder_o_reg        <= 'b0;
        encoder_o_reg0       <= 'b0;
          enable_decoder_in    <= 'b0;
        enable_encoder_i_reg <= 'b0;
        word_ct              <= 'b0;
       end
       else begin
          enable_encoder_i_reg <= enable_encoder_i;
          enable_decoder_in    <= valid_encoder_o;
//          encoder_o_reg        <= 'b0;
          if(word_ct[0]) error_counter          <= $random;
          word_ct              <= word_ct + 1;
// bit error injection in encoder_o_reg
          encoder_i_reg     <= encoder_i;
          encoder_o_reg0    <= encoder_o;
//      error_counterQ    <= error_counter;
          if(error_counter[N-1:0]==1) begin //
           err_inj[1]        <= error_counter[29];
             encoder_o_reg  <= encoder_o^err_inj;    // inject bad bits
          end
          else begin              // clean version
             encoder_o_reg  <= encoder_o;
             err_inj        <= 2'b0;
       end
         if(word_ct<256) begin
           bad_bit_ct  <= bad_bit_ct + (encoder_o_reg0[1]^encoder_o_reg[1])
                        + (encoder_o_reg0[0]^encoder_o_reg[0]);
         $display("error_counter,err_inj = %h %b %d %d",
                 error_counter,err_inj,bad_bit_ct,word_ct);
          end
       end

// insert your convolutional encoder here
  encoder encoder1       (
      .clk,
      .rst,
      .enable_i(enable_encoder_i), //_reg),
      .d_in     (encoder_i),       //_reg),
      .valid_o (valid_encoder_o),
      .d_out    (encoder_o)    );

// insert your term project code here
  decoder decoder1       (
      .clk,
      .rst,
      .enable (enable_decoder_in),
      .d_in   (encoder_o_reg),
      .d_out  (decoder_o)    );

endmodule
```

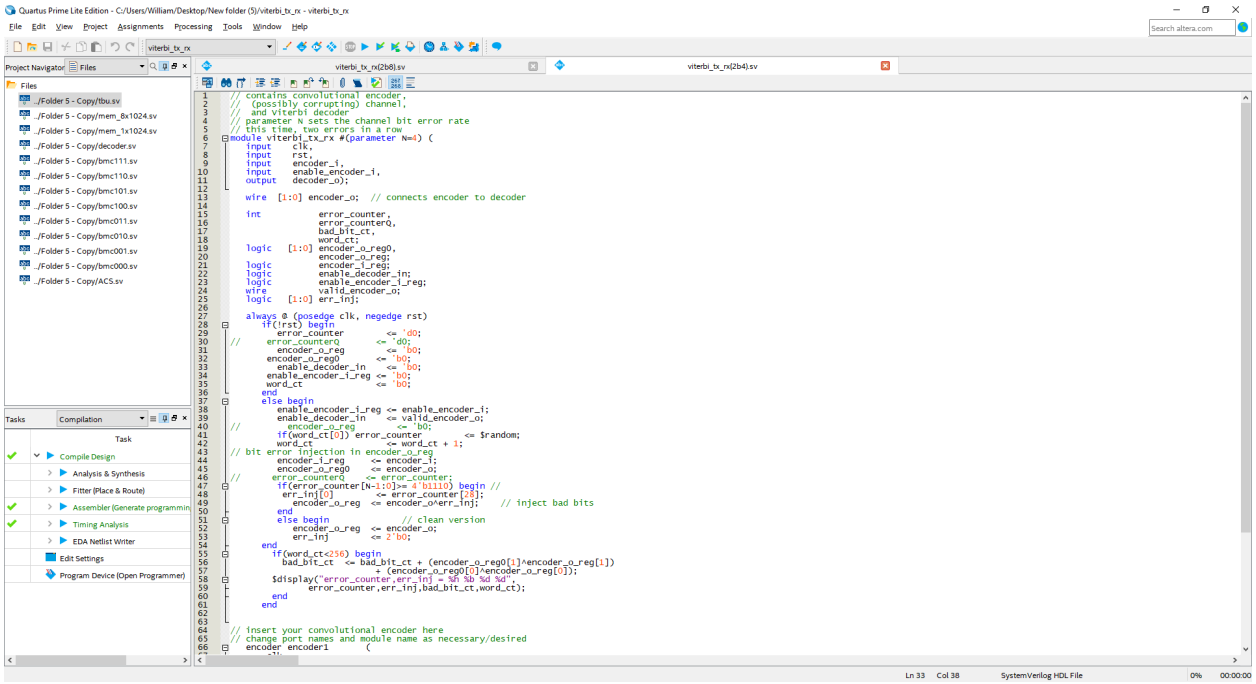
```
# good =          256, bad =            0,          8 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

2b3.

```
# good =          256, bad =            0,              5 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

```
// contains convolutional encoder,
//  (possibly corrupting) channel,
//  and viterbi decoder
// parameter N sets the channel bit error rate
// this time, two errors in a row
module viterbi_tx_rx #(parameter N=4) (
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;  // connects encoder to decoder

    int           error_counter,
                  error_counterQ,
                  bad_bit_ct,
                  word_ct;
    logic   [1:0] encoder_o_reg0,
                  encoder_o_reg;
    logic         encoder_i_reg;
    logic         enable_decoder_in;
    logic         enable_encoder_i_reg;
    wire          valid_encoder_o;
    logic   [1:0] err_inj;

    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter        <= 'd0;
//      error_counterQ       <= 'd0;
          encoder_o_reg        <= 'b0;
        encoder_o_reg0       <= 'b0;
          enable_decoder_in    <= 'b0;
        enable_encoder_i_reg <= 'b0;
        word_ct              <= 'b0;
       end
       else begin
          enable_encoder_i_reg <= enable_encoder_i;
          enable_decoder_in    <= valid_encoder_o;
//          encoder_o_reg        <= 'b0;
          if(word_ct[0]) error_counter        <= $random;
        word_ct                <= word_ct + 1;
// bit error injection in encoder_o_reg
          encoder_i_reg     <= encoder_i;
          encoder_o_reg0    <= encoder_o;
//      error_counterQ    <= error_counter;
          if(error_counter[N-1:0]=='1) begin //
           err_inj        <= error_counter[29:28];
             encoder_o_reg  <= encoder_o^err_inj;    // inject bad bits
          end
          else begin               // clean version
             encoder_o_reg  <= encoder_o;
             err_inj        <= 2'b0;
       end
         if(word_ct<256) begin
           bad_bit_ct  <= bad_bit_ct + (encoder_o_reg0[1]^encoder_o_reg[1])
                        + (encoder_o_reg0[0]^encoder_o_reg[0]);
         $display("error_counter,err_inj = %h %b %d %d",
                 error_counter,err_inj,bad_bit_ct,word_ct);
         end
       end

// insert your convolutional encoder here
// change port names and module name as necessary/desired
   encoder encoder1       (
       .clk,
       .rst,
       .enable_i(enable_encoder_i), //_reg),
       .d_in     (encoder_i),       //_reg),
       .valid_o (valid_encoder_o),
       .d_out   (encoder_o)   );

// insert your term project code here
   decoder decoder1       (
       .clk,
       .rst,
       .enable (enable_decoder_in),
       .d_in   (encoder_o_reg),
       .d_out  (decoder_o)   );

endmodule
```

2b4.




```
// insert your convolutional encoder here
// change port names and module name as necessary/desired
   encoder encoder1       (
       .clk,
       .rst,
       .enable_i(enable_encoder_i), //_reg),
       .d_in    (encoder_i),        //_reg),
       .valid_o (valid_encoder_o),
       .d_out   (encoder_o)   );

// insert your term project code here
   decoder decoder1       (
       .clk,
       .rst,
       .enable (enable_decoder_in),
       .d_in   (encoder_o_reg),
       .d_out  (decoder_o)   );

endmodule
```

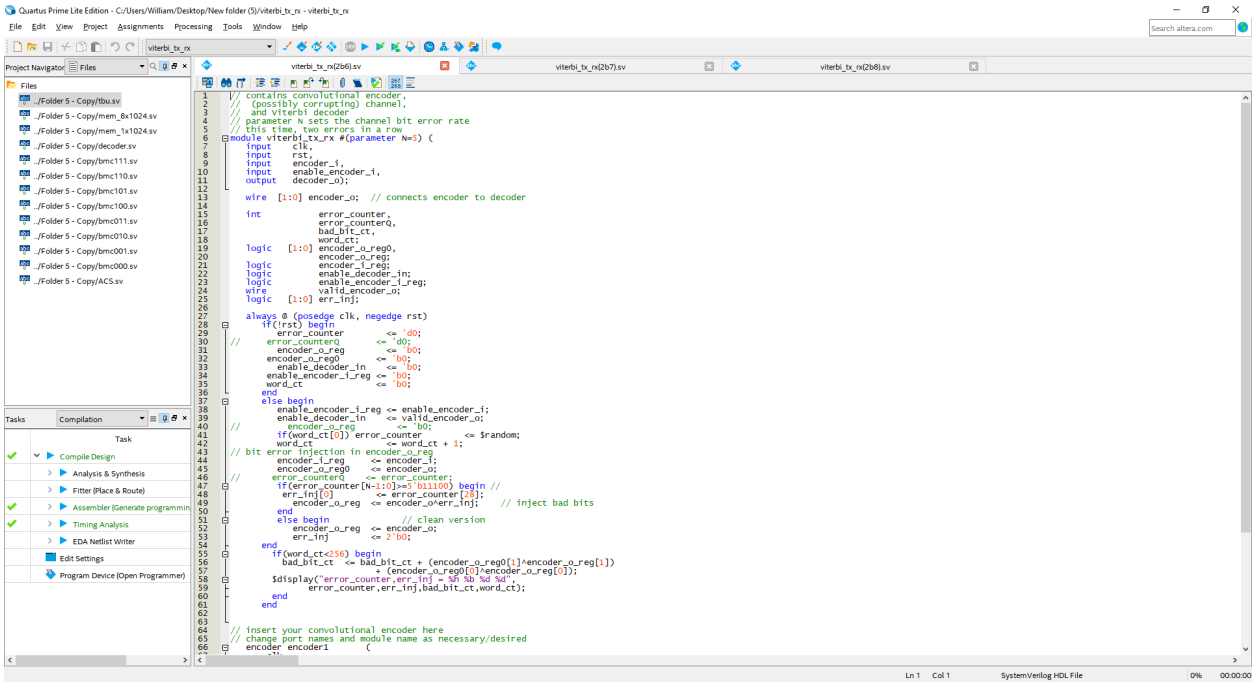
```
# good =          256, bad =            0,              4 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

2b5.

```
// contains convolutional encoder,
//  (possibly corrupting) channel,
//  and viterbi decoder
// parameter N sets the channel bit error rate
// this time, two errors in a row
module viterbi_tx_rx #(parameter N=4) (
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;  // connects encoder to decoder

    int            error_counter,
                   error_counterQ,
                   bad_bit_ct,
                   word_ct;
    logic   [1:0] encoder_o_reg0,
                   encoder_o_reg;
    logic          encoder_i_reg;
    logic          enable_decoder_in;
    logic          enable_encoder_i_reg;
    wire           valid_encoder_o;
    logic   [1:0] err_inj;

    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter         <= 'd0;
//        error_counterQ        <= 'd0;
          encoder_o_reg         <= 'b0;
        encoder_o_reg0          <= 'b0;
          enable_decoder_in     <= 'b0;
        enable_encoder_i_reg <= 'b0;
        word_ct                 <= 'b0;
       end
       else begin
          enable_encoder_i_reg <= enable_encoder_i;
          enable_decoder_in    <= valid_encoder_o;
//          encoder_o_reg        <= 'b0;
          if(word_ct[0]) error_counter        <= $random;
          word_ct                <= word_ct + 1;
// bit error injection in encoder_o_reg
          encoder_i_reg      <= encoder_i;
          encoder_o_reg0     <= encoder_o;
//        error_counterQ     <= error_counter;
          if(error_counter[N-1:0]>=4'b1110) begin //
            err_inj[1]       <= error_counter[29];
              encoder_o_reg  <= encoder_o^err_inj;    // inject bad bits
          end
          else begin               // clean version
              encoder_o_reg  <= encoder_o;
              err_inj        <= 2'b0;
       end
          if(word_ct<256) begin
            bad_bit_ct  <= bad_bit_ct + (encoder_o_reg0[1]^encoder_o_reg[1])
                                      + (encoder_o_reg0[0]^encoder_o_reg[0]);
        $display("error_counter,err_inj = %h %b %d %d",
                 error_counter,err_inj,bad_bit_ct,word_ct);
          end
       end

// insert your convolutional encoder here
// change port names and module name as necessary/desired
   encoder encoder1        (
      .clk,
      .rst,
      .enable_i(enable_encoder_i), //_reg),
      .d_in    (encoder_i),        //_reg),
      .valid_o (valid_encoder_o),
      .d_out   (encoder_o)   );

// insert your term project code here
   decoder decoder1       (
      .clk,
      .rst,
      .enable (enable_decoder_in),
      .d_in   (encoder_o_reg),
      .d_out  (decoder_o)   );

endmodule
```

```
# good =          256, bad =            0,              6 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

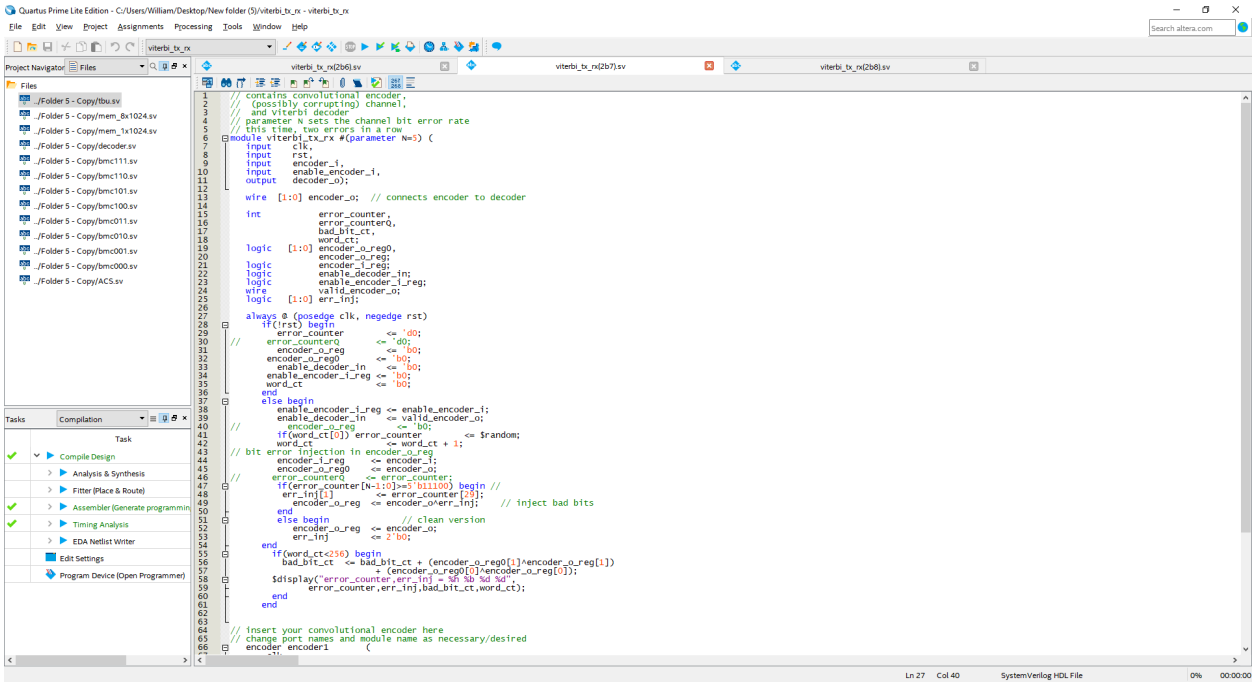
2b6.




```
// insert your convolutional encoder here
// change port names and module name as necessary/desired
  encoder encoder1       (
      .clk,
      .rst,
      .enable_i(enable_encoder_i), //_reg),
      .d_in     (encoder_i),       //_reg),
      .valid_o (valid_encoder_o),
      .d_out    (encoder_o)    );

// insert your term project code here
  decoder decoder1       (
      .clk,
      .rst,
      .enable (enable_decoder_in),
      .d_in   (encoder_o_reg),
      .d_out  (decoder_o)    );

endmodule
```

```
# good =          256, bad =            0,              7 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

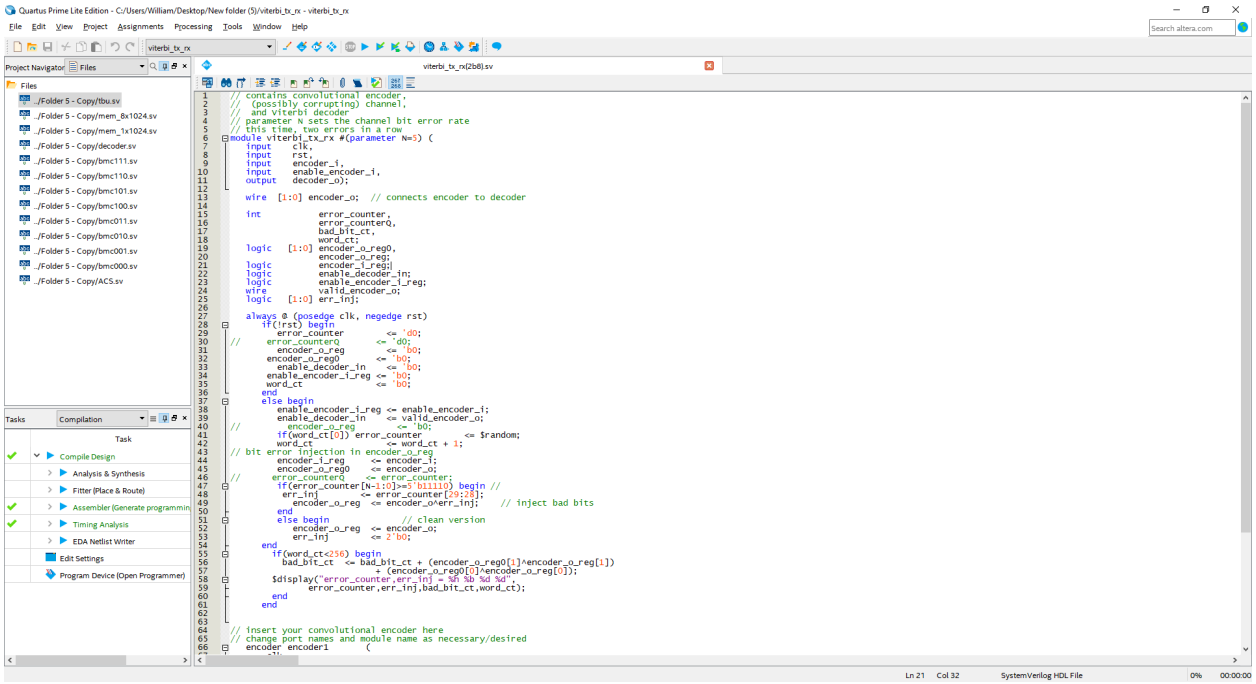
2b7.




```
// insert your convolutional encoder here
// change port names and module name as necessary/desired
encoder encoder1      (
    .clk,
    .rst,
    .enable_i(enable_encoder_i), //_reg),
    .d_in    (encoder_i),        //_reg),
    .valid_o (valid_encoder_o),
    .d_out   (encoder_o)   );

// insert your term project code here
decoder decoder1      (
    .clk,
    .rst,
    .enable (enable_decoder_in),
    .d_in    (encoder_o_reg),
    .d_out  (decoder_o)   );

endmodule
```

```
# good =          256, bad =            0,              8 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

2b8.




```
// insert your convolutional encoder here
// change port names and module name as necessary/desired
  encoder encoder1      (
      .clk,
      .rst,
      .enable_i(enable_encoder_i), //_reg),
      .d_in     (encoder_i),       //_reg),
      .valid_o (valid_encoder_o),
      .d_out    (encoder_o)   );

// insert your term project code here
  decoder decoder1      (
      .clk,
      .rst,
      .enable (enable_decoder_in),
      .d_in   (encoder_o_reg),
      .d_out  (decoder_o)   );

endmodule
```

```
# good =          256, bad =            0,              5 bad_bits
# ** Note: $stop    : H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv(217)
#    Time: 1045 ns  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at H:/Folder 5 - Copy/viterbi_tx_rx_tb (2b).sv line 217
```

Part C

Testbench: same as part A

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic       enable_decoder_in;
    wire        valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 1;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
           error_counter  <= 4'd0;
           encoder_o_reg  <= 2'b00;
           enable_decoder_in <= 1'b0;
           error_bit8 <= 3'd0;
           error_bit32 <= 5'd0;
            word_ct = 0;
       end
       else     begin
           enable_decoder_in <= valid_encoder_o;
           encoder_o_reg  <= 2'b00;
           error_counter  <= error_counter + 4'd1;
           error_bit8 <= error_bit8 + 3'd1;
           error_bit32 <= error_bit32 + 5'd1;
            word_ct              <= word_ct + 1;


           if(error_counter == 4'b1001 || error_counter == 4'b1000 || error_counter == 4'b0111 || error_counter == 4'b0110 ) begin
           //2c //with 3 consecutive # good =256, bad=0 // with 4 consecutive  good =     181, bad =          75 BROKE THE CODE
              encoder_o_reg  <= {encoder_o[1],~encoder_o[0]};  // inject 4 bad bit out of every 16
                if(word_ct<256)begin
                   bad_bit_ct      <= bad_bit_ct + 1;
                   $display("error_counter = %d",bad_bit_ct);
             end
         end


           else
              encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

 // insert your convolutional encoder here
 // change port names and module name as necessary/desired
    encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );

 // insert your term project code here
    decoder decoder1       (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

 endmodule
```

With 3 consecutive, we got 0 bad bits and 4 consecutive broke our code.

```
# error_counter =          62
# error_counter =          63
# error_counter =          64
```

```
# boo! in = 1, out = 0
# good =         181, bad =          75
# ** Note: $stop    : C:/Users/William/Desktop/Folder 5/viterbi_tx_rx_tb(4).sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at C:/Users/William/Desktop/Folder 5/viterbi_tx_rx_tb(4).sv line 371

VSIM 6>
```

Part D

Testbench: same as part A

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic        enable_decoder_in;
    wire         valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 1;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter  <= 4'd0;
          encoder_o_reg  <= 2'b00;
          enable_decoder_in <= 1'b0;
          error_bit8 <= 3'd0;
          error_bit32 <= 5'd0;
           word_ct = 0;
       end
       else     begin
          enable_decoder_in <= valid_encoder_o;
          encoder_o_reg  <= 2'b00;
          error_counter  <= error_counter + 4'd1;
          error_bit8 <= error_bit8 + 3'd1;
          error_bit32 <= error_bit32 + 5'd1;
           word_ct                <= word_ct + 1;


         if(error_counter == 4'b1001 || error_counter == 4'b1000 || error_counter == 4'b0111 || error_counter == 4'b0110 ) begin
          //2d //with 3 consecutive # good =256, bad=0 // with 4 consecutive  good =     226, bad =     30 BROKE THE CODE
             encoder_o_reg  <= {~encoder_o[1],encoder_o[0]};  // inject 4 bits every 16
                if(word_ct<256)begin
                   bad_bit_ct      <= bad_bit_ct + 1;
                   $display("error_counter = %d",bad_bit_ct);
             end
         end

          else
             encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
       end

 // insert your convolutional encoder here
 // change port names and module name as necessary/desired
    encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );
```

With 3 consecutive we got 0 bad bits, and with 4 consecutive, it broke our code.

```
// insert your term project code here
   decoder decoder1       (
      .clk(clk),
      .rst(rst),
      .enable(enable_decoder_in),
      .d_in(encoder_o_reg),
      .d_out(decoder_o)
      );

 endmodule
```

```
# error_counter =          61
# error_counter =          62
# error_counter =          63
# error_counter =          64
```

```
# boo! in = 1, out = 0
# good =         226, bad =          30
# ** Note: $stop    : C:/Users/William/Desktop/Folder 5/viterbi_tx_rx_tb(4).sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at C:/Users/William/Desktop/Folder 5/viterbi_tx_rx_tb(4).sv line 371
```

Part E

Testbench: same as part A

```
module viterbi_tx_rx(
    input    clk,
    input    rst,
    input    encoder_i,
    input    enable_encoder_i,
    output   decoder_o);

    wire  [1:0] encoder_o;

    logic   [3:0] error_counter;
    logic   [1:0] encoder_o_reg;

    logic       enable_decoder_in;
    wire        valid_encoder_o;

    //Logic for error for bit 0 and 1
    logic [2:0] error_bit8;
    logic [4:0] error_bit32;
    int bad_bit_ct = 2;
    int word_ct;


    always @ (posedge clk, negedge rst)
       if(!rst) begin
          error_counter  <= 4'd0;
          encoder_o_reg  <= 2'b00;
          enable_decoder_in <= 1'b0;
          error_bit8 <= 3'd0;
          error_bit32 <= 5'd0;
           word_ct = 0;
      end
      else     begin
          enable_decoder_in <= valid_encoder_o;
          encoder_o_reg  <= 2'b00;
          error_counter  <= error_counter + 4'd1;
          error_bit8 <= error_bit8 + 3'd1;
          error_bit32 <= error_bit32 + 5'd1;
           word_ct              <= word_ct + 1;


               if(error_counter == 4'b1001 || error_counter == 4'b1000 )begin
               //2e # with 2 injected every 16 good =     256, bad =     0 , Injected = 32 // with 4 consecutive,  good =   211, bad =    45   BROKE THE CODE
            encoder_o_reg  <= {~encoder_o[1],~encoder_o[0]};
               if(word_ct<256)begin
                  bad_bit_ct     <= bad_bit_ct + 2;
                   $display("error_counter = %d",bad_bit_ct);
             end
          end

          else
             encoder_o_reg  <= {encoder_o[1],encoder_o[0]};
      end

 // insert your convolutional encoder here
 // change port names and module name as necessary/desired
   encoder encoder1       (
       .clk(clk),
       .rst(rst),
       .enable_i(enable_encoder_i),
       .d_in(encoder_i),
       .valid_o(valid_encoder_o),
       .d_out(encoder_o)   );

 // insert your term project code here
```

With 2 bits injected every 16, we got 0 bad bits. With 4 bad bits injected in a row, it broke the code.

```
 // insert your term project code here
   decoder decoder1       (
       .clk(clk),
       .rst(rst),
       .enable(enable_decoder_in),
       .d_in(encoder_o_reg),
       .d_out(decoder_o)
       );

 endmodule
```

```
# error_counter =          22
# error_counter =          24
# error_counter =          26
# error_counter =          28
# error_counter =          30
# error_counter =          32
# error_counter =          34
# error_counter =          36
# error_counter =          38
# error_counter =          40
# error_counter =          42
# error_counter =          44
# error_counter =          46
# error_counter =          48
# error_counter =          50
# error_counter =          52
# error_counter =          54
# error_counter =          56
# error_counter =          58
# error_counter =          60
# error_counter =          62
# error_counter =          64
```

```
# yaa! in = 1, out = 1
# good =         211, bad =          45
# ** Note: $stop    : C:/Users/William/Downloads/viterbi_tx_rx_tb(4).sv(371)
#    Time: 1381700 ps  Iteration: 0  Instance: /viterbi_tx_rx_tb
# Break in Module viterbi_tx_rx_tb at C:/Users/William/Downloads/viterbi_tx_rx_tb(4).sv line 371

VSIM 12>
```

Final Results of all Test

2				NonRandom Tests			
3	Test Number	ViterbiTxRx	BER	SymbolPattern	BitPattern	BadBitsIn	BadBitsOut
4	2.a.1	viterbi_tx_rx(a1).sv	1/8	1 in a row	01	32	0
5	2.a.2	viterbi_tx_rx(a2).sv	1/8	1 in a row	10	32	0
6	2.a.3	viterbi_tx_rx(a3).sv	2/8	1 in a row	11	64	51
7	2.a.4	viterbi_tx_rx(a4).sv	2/16	2 in a row	01	32	0
8	2.a.5	viterbi_tx_rx(a5).sv	2/16	2 in a row	10	32	0
9	2.a.6	viterbi_tx_rx(a6).sv	4/32	4 in a row	01	32	33
10	2.a.7	viterbi_tx_rx(a7).sv	4/32	4 in a row	10	32	16
11	2.a.8	viterbi_tx_rx(a8).sv	2/32	2 in a row	11	32	24
12	2.C.	viterbi_tx_rx(2c).sv	2/16	1 in a row	01	64	75
13	2.D	viterbi_tx_rx(2d).sv	4/16	1 in a row	10	64	30
14	2.E	viterbi_tx_rx(2e).sv	2/16	2 in a row	11	64	45
15							
16							
17							
18							
19							
20							
21							
22							
23							
24							
25							
26				Random Tests			
27	Test Number	ViterbiTxRx	Avg. BER	SymbolPattern	BitPattern	BadBitsIn	BadBitsOut
28	2.b.1	viterbi_tx_rx(2b1).sv	3/128	1 in a row	01	6	0
29	2.b.2	viterbi_tx_rx(2b2).sv	1/32	1 in a row	10	8	0
30	2.b.3	viterbi_tx_rx(2b3).sv	5/256	1 in a row	11	5	0
31	2.b.4	viterbi_tx_rx(2b4).sv	1/64	2 in a row	01	4	0
32	2.b.5	viterbi_tx_rx(2b5).sv	3/128	2 in a row	10	6	0
33	2.b.6	viterbi_tx_rx(2b6).sv	7/256	4 in a row	01	7	0
34	2.b.7	viterbi_tx_rx(2b7).sv	1/32	4 in a row	10	8	0
35	2.b.8	viterbi_tx_rx(2b8).sv	5/256	2 in a row	11	5	0

The decoder is performing well because we expect badbitsout to be significantly lower than badbitsin. The tests are categorized into nonrandom and random, each presenting different conditions and bit patterns. Variations in performance across tests suggest the decoder's error correction efficiency.